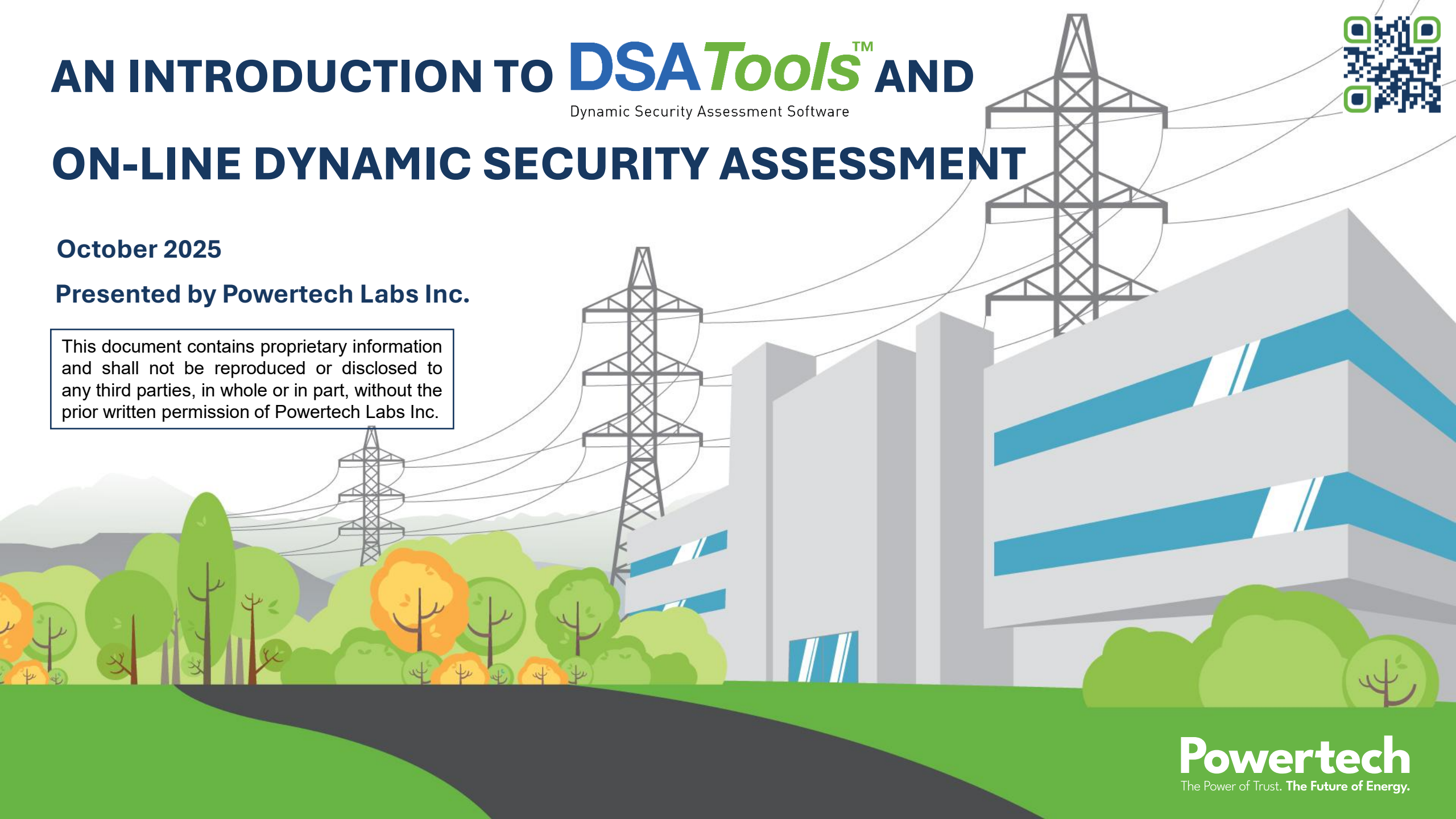
AN INTRODUCTION TO DSATools™ AND ON-LINE DYNAMIC SECURITY ASSESSMENT

Dynamic Security Assessment Software



October 2025

Presented by Powertech Labs Inc.

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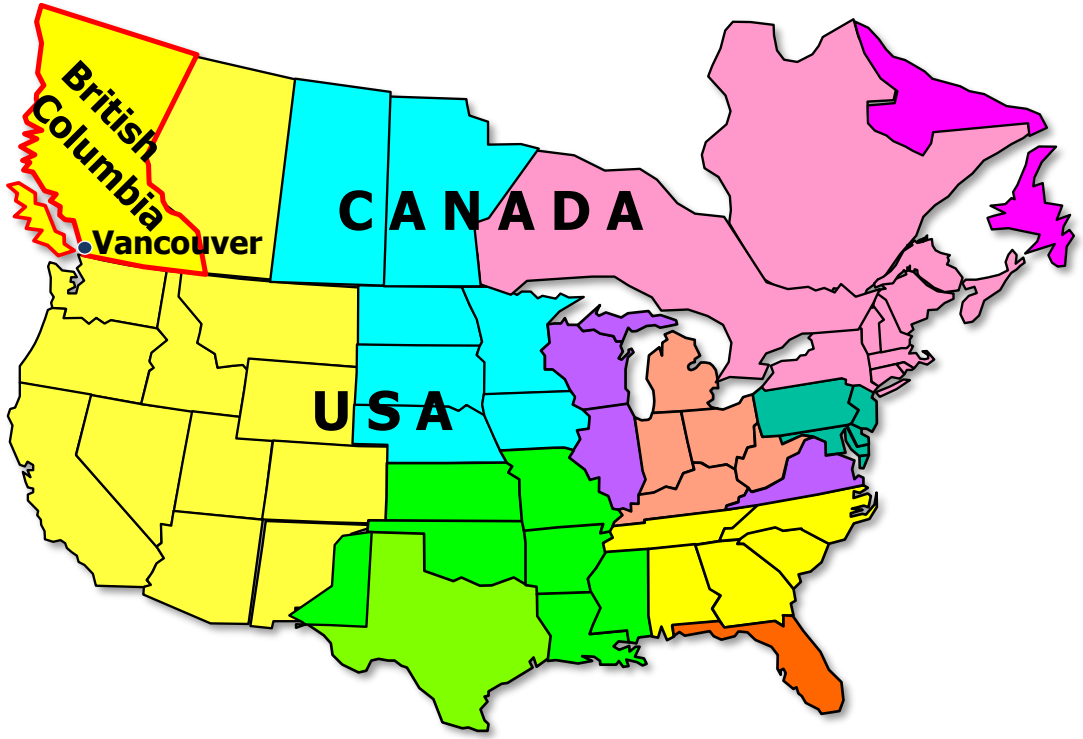
Powertech

The Power of Trust. **The Future of Energy.**

Powertech WHO ARE WE?

The Power of Trust. The Future of Energy.

- Founded in 1989, and located in the great **Vancouver** area, **BC**, **Canada**
- Subsidiary of British Columbia Hydro and Power Authority
- Large multidisciplinary laboratory, 11-acre site with **34 primary labs**
- Over **40 years** of specialized engineering experience
- **250+** employees, technologists, professional engineers, and PhDs
- Main business includes various types of **testing** (high voltage/power, hydrogen, and many more), **engineering consulting**, and **software development**
- Management system registered to **ISO 27001**, ISO 9001, ISO 14001, and ISO 45001





OVERVIEW

- An introduction to DSATools™
- On-line Dynamic Security Assessment (DSA) Overview
- On-line DSA Application Examples
- DSA System Demonstration

AN INTRODUCTION TO DSATOOLS™

DSATOOLS – BACKGROUND

- A suite of power system analysis software programs, originally developed in mid-1990s
- 25+ years history of **online** and offline applications
- **Version 25** was released July 2025
- Powertech provides comprehensive user support and services, including
  - Software upgrades
  - Software customizations
  - Model development
  - Technical support
  - Training



DSATOOLS – CORE MODULES

PSAT – Powerflow and Short-circuit Analysis Tool

VSAT – Voltage Security Assessment Tool

TSAT – Transient Security Assessment Tool

SSAT – Small Signal Analysis Tool

DSA Manager – On-line DSA interface module

DSATOOLS ADD-ON MODULES

UDM Editor – User-defined model editor

DSAOA – Output analysis for VSAT and TSAT

ePMU – Simulated PMU data generator

CDT – Control design toolbox for PSS design

SSR – Sub-synchronous resonance analysis

TRI – TSAT-RTDS™ Interface for hybrid simulation

TPI – TSAT-PSCAD™ Interface for hybrid simulation

ST DSA Manager – Study mode manager for on-line DSA

LA DSA Manager – Look Ahead manager for on-line DSA

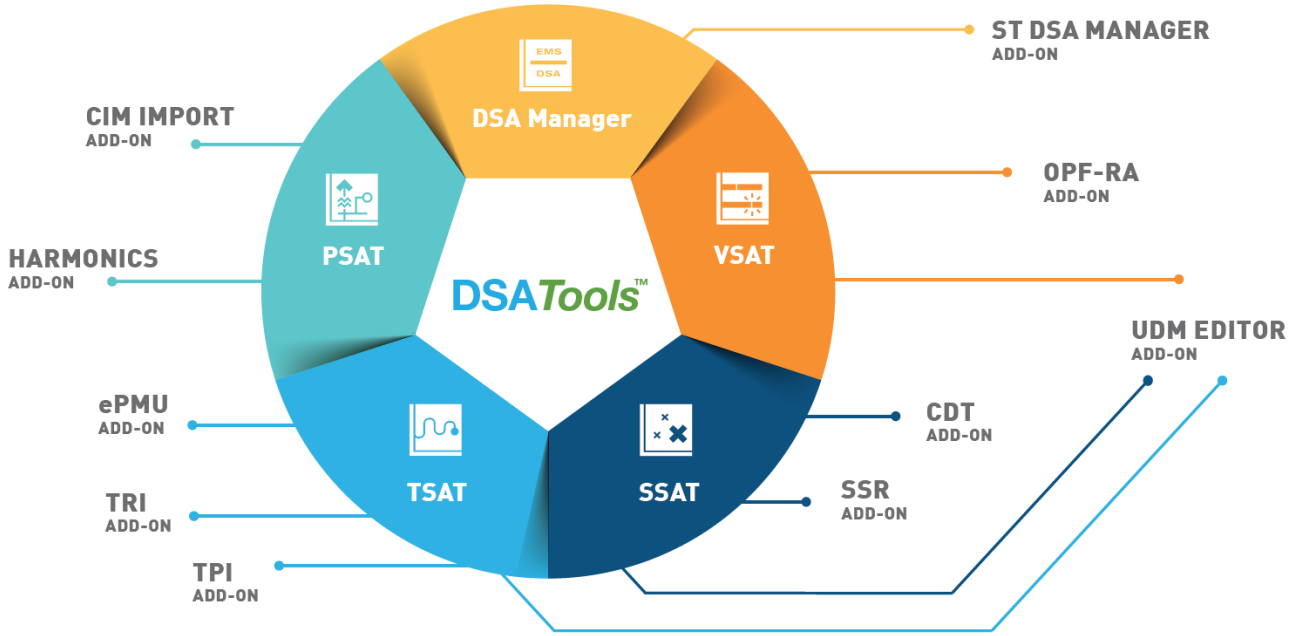
Harmonics – Harmonic analysis

CIM Import – CIM data importer

OPF-RA – Remedial action determination using OPF

DSATOOLS – APPLICATIONS

- Power system analysis
  - Planning, operation, and many other type of studies
  - On-line Dynamic Security Assessment (DSA)
  - Teaching and research
- Used by
  - Regulators
  - Independent System Operators (ISO)
  - Power market operators
  - Utilities (including transmission, generation, and distribution companies)
  - Manufacturers and vendors
  - Consultants
  - Research institutes and universities



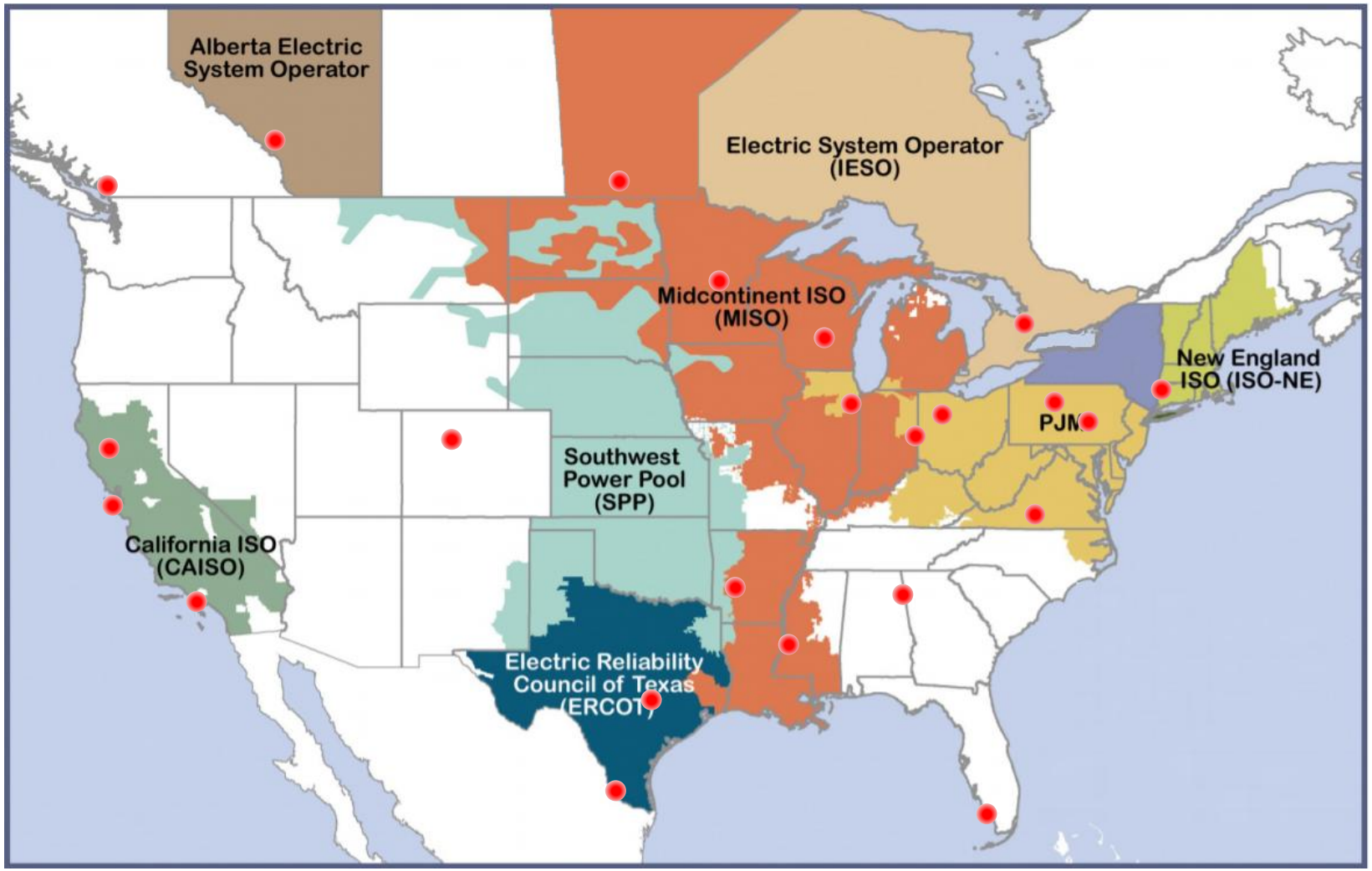
WORLDWIDE DSATOOLS™ USERS

- Worldwide client base



ON-LINE DSATOOLS™ APPLICATIONS IN NORTH AMERICA

- On-line DSA installations in North America – including 8 ISOs and major utilities



SELECTED ON-LINE DSATOOLS™ USERS - WORLDWIDE

	Licensed Products	Year of Licensing
AEMO (Australia Energy Market Operator, Australia)	VSAT	2008
KPX (Korea Power Exchange, South Korea)	VSAT/TSAT/SSAT	2012/2016
EirGrid (Ireland)	VSAT/TSAT	2002
EMA (Singapore)	VSAT/TSAT	2012
JEPC (State Grid Jiangsu Electric Power Company, China)	TSAT/VSAT	2018/2019
KEPRI (KEPCO Research Institute, South Korea)	TSAT/VSAT/SSAT	2018
National Grid (Saudi Arabia)	VSAT/TSAT	2014
National Grid ESO (UK)	TSAT/VSAT	2012/2022
NTDC (Pakistan)	VSAT/TSAT	2022
SEB (Sarawak Energy Berhad, Malaysia)	VSAT/TSAT	2012
SONELGAZ (Algeria)	VSAT/TSAT	2022
Statnett (Norway)	VSAT/TSAT	2011
TNB (Tenaga Nasional Berhad, Malaysia)	VSAT/TSAT	2005
Transelectric (Ecuador)	VSAT/TSAT	2015
Transpower (New Zealand)	VSAT/TSAT	2005/2010/2013
XM (Colombia)	VSAT	2010

PRIMARY FEATURES

- Analysis methods
  - **Powerflow:** Newton-Raphson, fast decoupled, short-circuit analysis, static network reduction, harmonic analysis
  - **Voltage stability:** P-V, Q-V, modal analysis, continuation powerflow, sensitivity analysis
  - **Transient stability:** time-domain simulations (Runge-Kutta, trapezoidal), transient stability index based on EEAC
  - **Small signal stability:** eigenvalue analysis methods suitable for different analysis objectives, frequency response calculations, sub-synchronous resonance analysis

PRIMARY FEATURES (CONT'D)

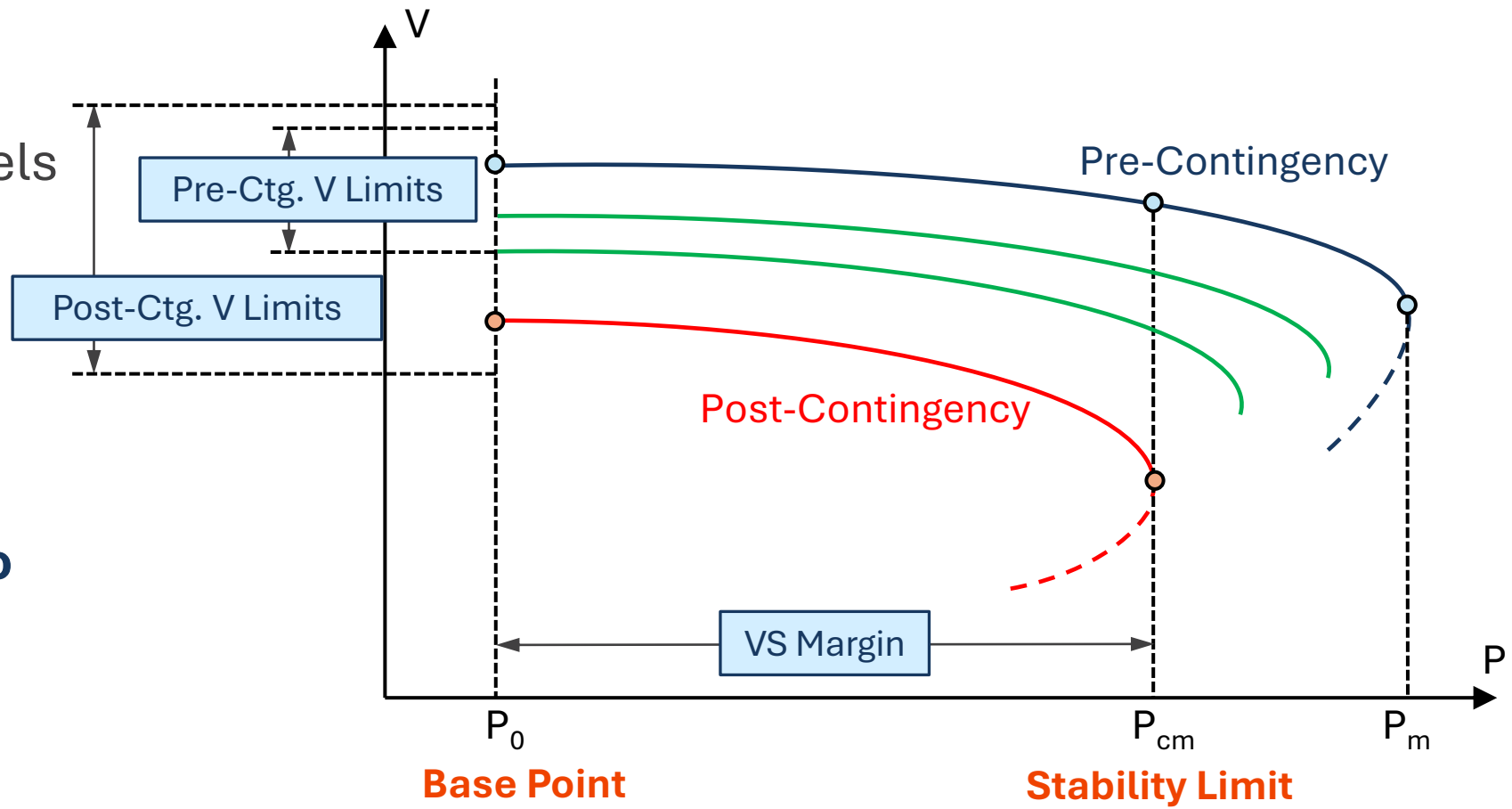
- Analysis of **multiple scenarios** and **multiple contingencies**
  - Data, model, analysis options can be customized and shared
  - Many options available to improve analysis efficiency/runtime and for logging and visualization of results
  - Use of security indices to quickly screen contingencies so detailed analysis can be done only for a small number of critical ones
  - Ideal for compliance studies such as NERC TPL and CIP-14

PRIMARY FEATURES (CONT'D)

- Comprehensive **modeling** support
- Options to generate **contingencies**
- **Transfer analysis** (**Limit Search**)
- For on-line dynamic security assessment (DSA)
  - Focus of this presentation, more details later
- Built-in **distributed computation** on multiple CPUs or CPU cores
  - This can significantly speed up the computations

SECURITY CRITERIA AVAILABLE

- Steady state
  - Thermal loading
  - Steady state voltage levels at pre- and post-contingency conditions
  - VAR reserve
  - **Weighted short-circuit ratio, short-circuit ratio**
  - Voltage stability



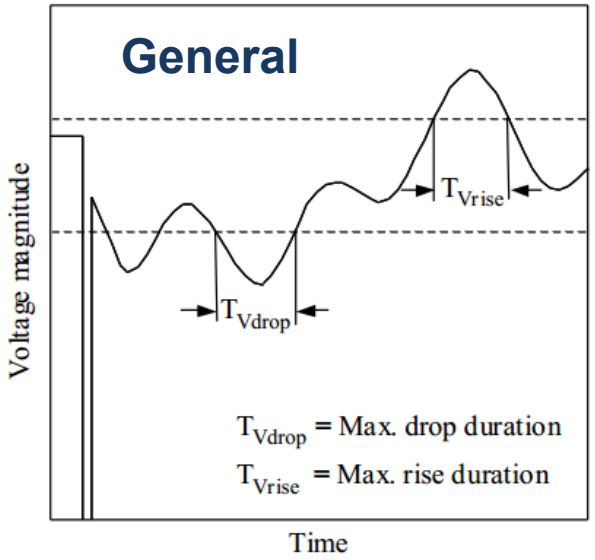
Definition of some steady state voltage security criteria

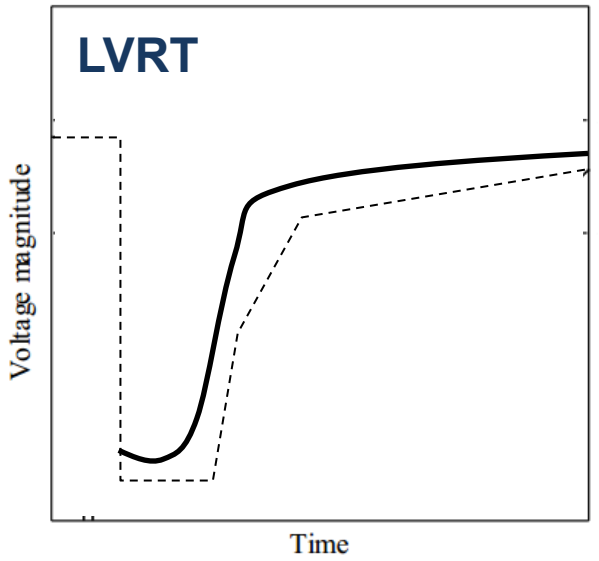
SECURITY CRITERIA AVAILABLE (CONT'D)

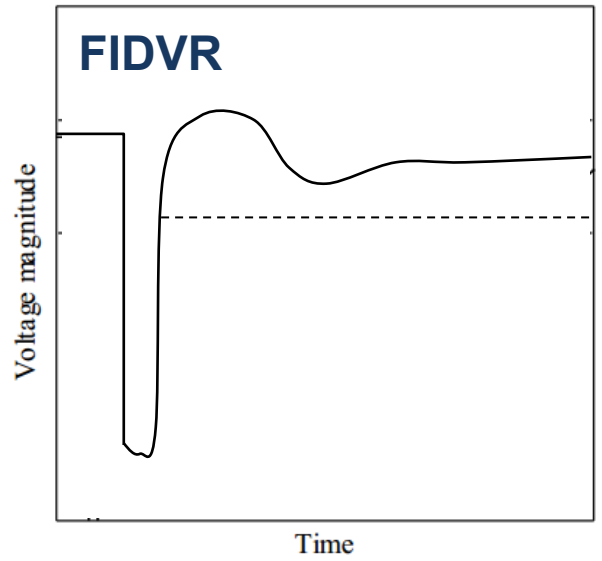
- Dynamic
  - Transient stability: may be based on
    - Power swing, or
    - Power angle
  - MW tripped due to out-of-step
  - Peak-to-peak angle separation
  - Torque on generator shaft
  - Critical clearance time (CCT) of faults
  - Damping factor of critical oscillatory modes
  - *…more on next page*

SECURITY CRITERIA AVAILABLE (CONT'D)

- Dynamic
  - Transient voltage performance
    - ❖ General voltage dip/rise
    - ❖ LVRT
    - ❖ FIDVR
  - Transient frequency performance: dip, rise, rate of change (ROCOF)
  - Relay margin







SECURITY CRITERIA APPLICATIONS

- Customizable for different study scenarios
- Can be applied to specific regions of a system
- Can be applied to specific contingencies
- Can be applied to determine transfer limits
- Fully support widely used study guides in North America
- **No lengthy post processing required – TSAT/VSAT/SSAT output detailed violation reports at the end of the computation**
- **Cascading analysis**

DYNAMIC MODELS

- Renewable generator models
  - 2nd generation (modular) model support
  - Applicable to wind, solar, and battery storage systems
- Custom relay and special protection system models
- User-defined models
  - FACTS
  - HVDC (LC and VSC)
  - Template models
- Load models
  - Composite load models

DYNAMIC MODELING

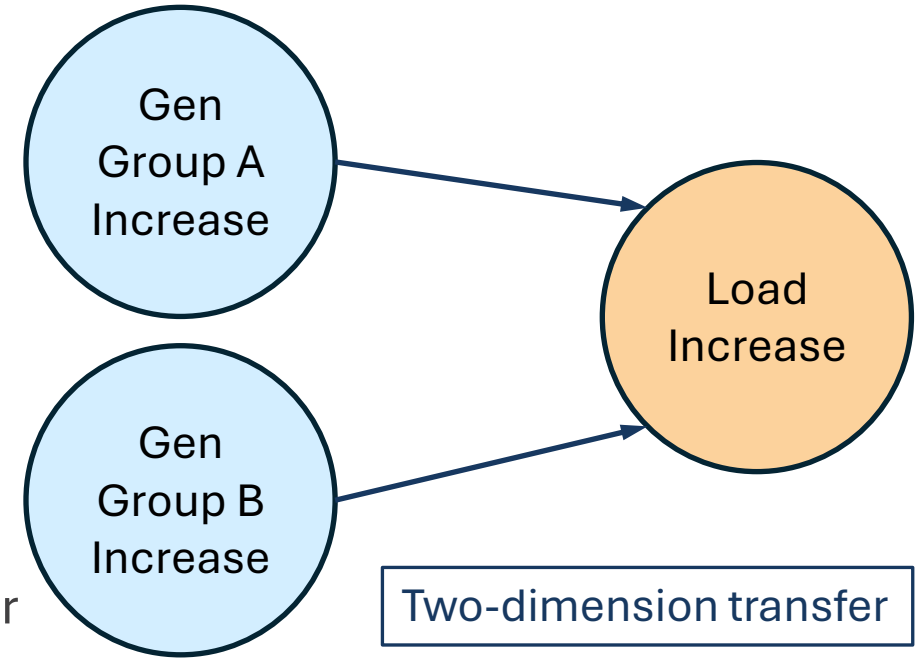
- Reads PSS/E and PSLF data directly (both powerflow and dynamics) or can use DSATools native format
- Most of the standard PSS/E and PSLF dynamic models are automatically converted
- Handling of “black-box”/proprietary models
  - Experience shows that the use of many black-box models can be problematic, suggest to avoid in favor of generic modeling
  - If must be used, TSAT/SSAT can convert some of the common black-box models
  - Powertech and other consultancies have worked with users to develop TSAT/SSAT compatible black-box models
  - **TSAT supports IEEE/CIGRE/IEC standard DLL API**
  - **ERCOT/MISO/SPP/PJM/Transpower require submission of TSAT compatible models**
  - **Australian TSOs/Saudi Electricity Company require submission of SSAT compatible models**

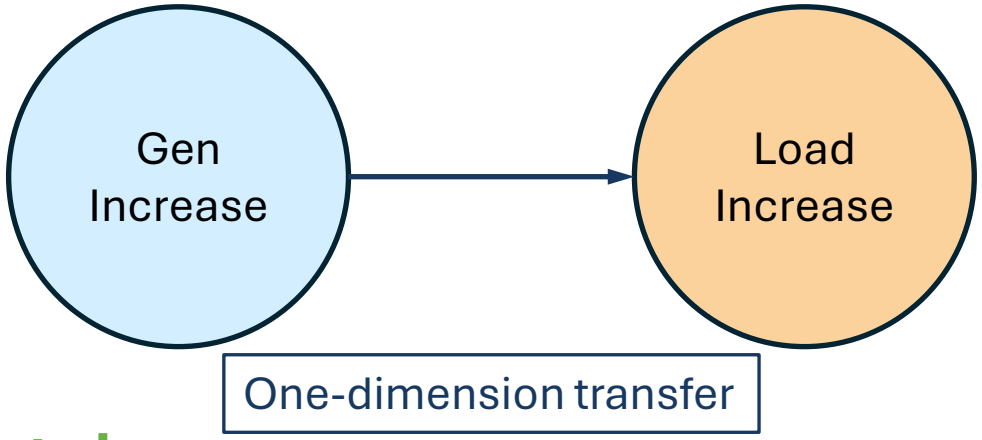
CONTINGENCIES

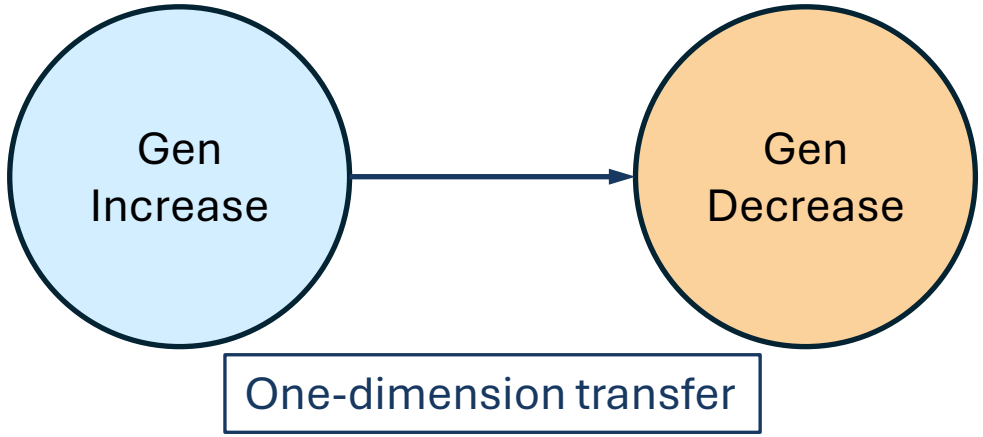
- A large variety of contingency types can be analyzed
- VSAT, TSAT and SSAT have features to generate contingencies using rules
  - For example, all N-1 outages for 500 kV system in areas x-y
  - VSAT also supports scripting to generate contingencies
- **Node-breaker modeling supported**, allowing special contingencies such as stuck breaker to be easily studied, and SPS scheme accurately modelled.

TRANSFER ANALYSIS

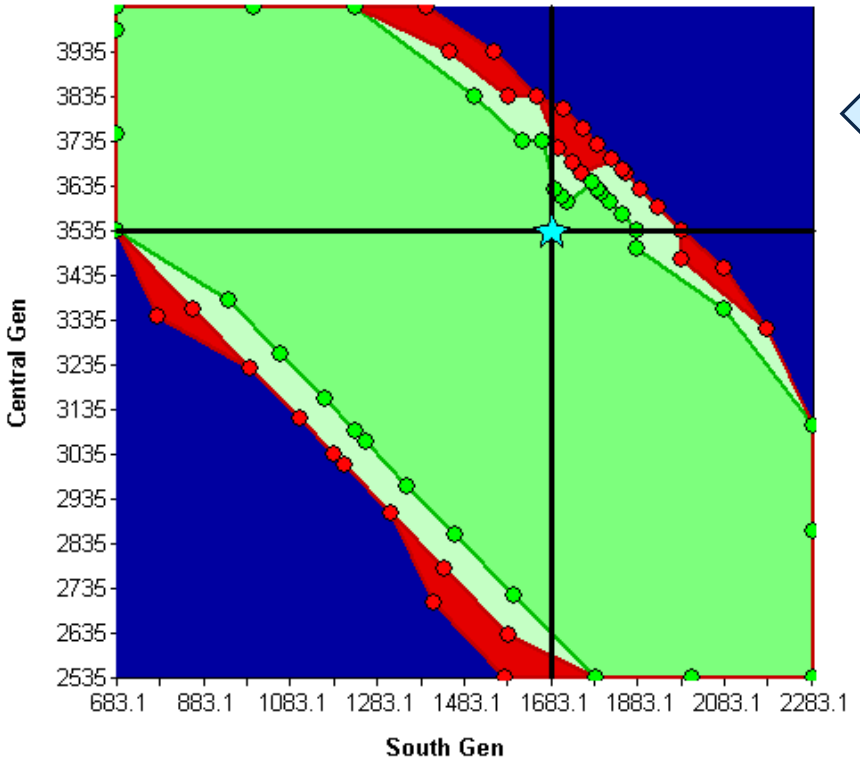
- Computation of **transfer (stability) limits**
  - Available in TSAT, VSAT & SSAT
  - Any supported security criteria can be considered
  - One- and two-dimensional limit computations
  - Flexible methods of dispatching system during a power transfer
  - Forward and backward limit search directions



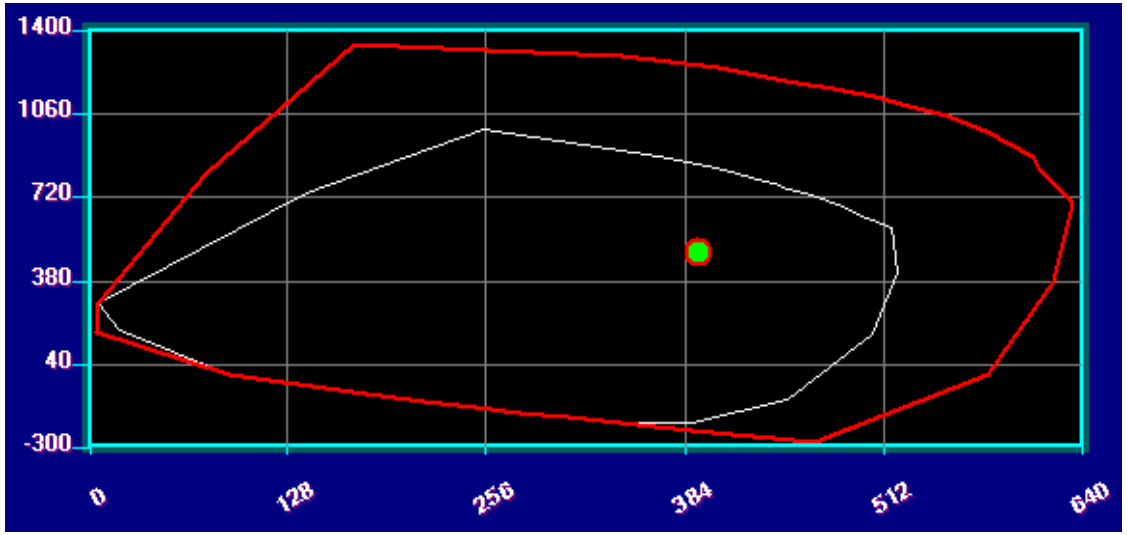




TRANSFER ANALYSIS EXAMPLES: 2-D TRANSFER NOMOGRAMS



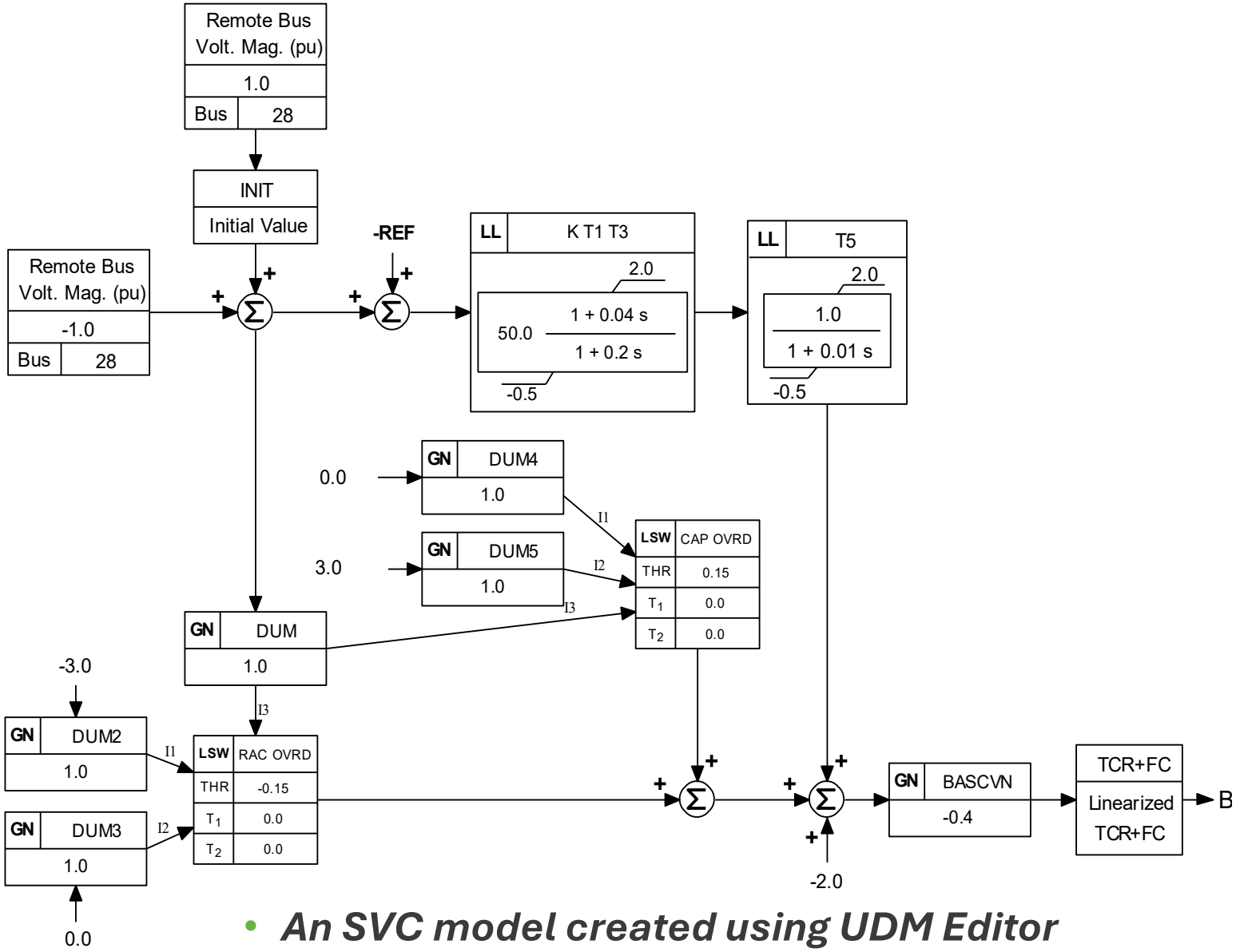
- Shows the secure/insecure regions, operating points, security violation types, and more
- Only shows the security boundaries



USER-DEFINED MODEL (UDM) EDITOR

- Drag & drop approach to create user-defined models using a graphical editor
- All user-defined model types are supported
  - Generator controls
  - Modularized renewable generators
  - FACTS
  - Special protection systems (SPS) models
  - HVDC models
  - Supervisory controllers
- Model example library provides a good starting point
- **Template models** can be created to use as parameterized “standard models”, allowing multiple instantiations and **directly reading of PSSE/PSLF dynamic data records**
- Built-in model validation within UDM Editor prior to use in studies

USER-DEFINED MODEL (UDM) EDITOR – EXAMPLE



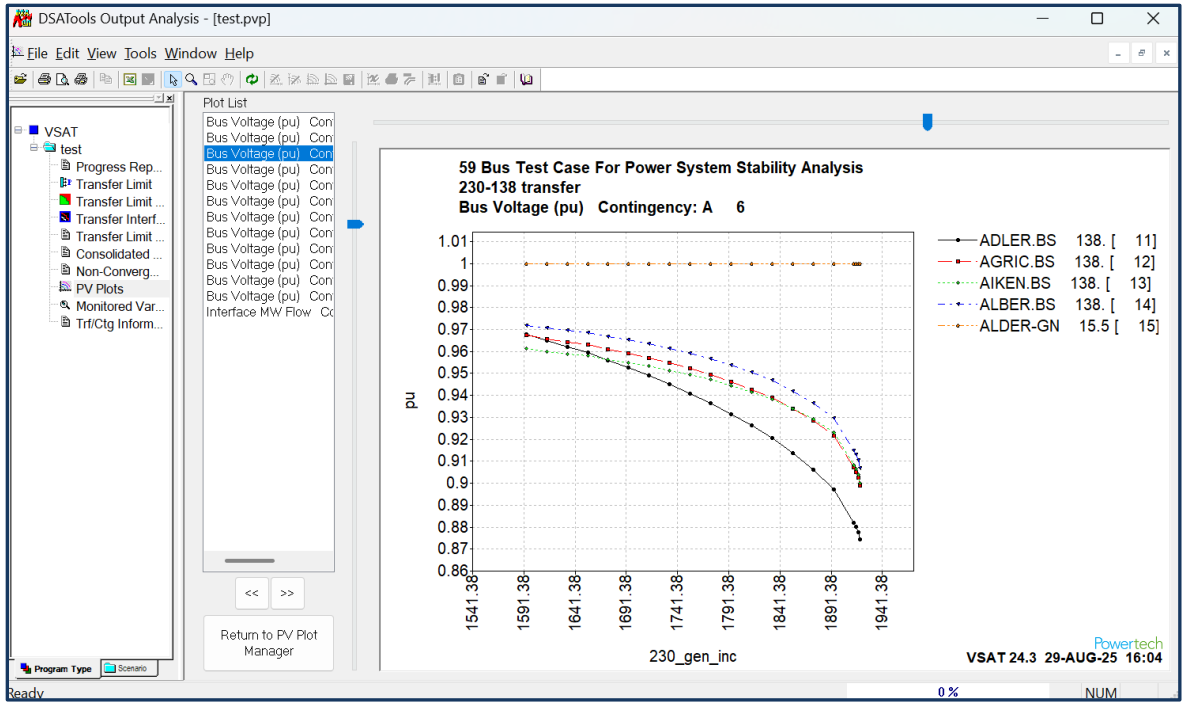
- ***An SVC model created using UDM Editor***

OUTPUT ANALYSIS – DSAOA

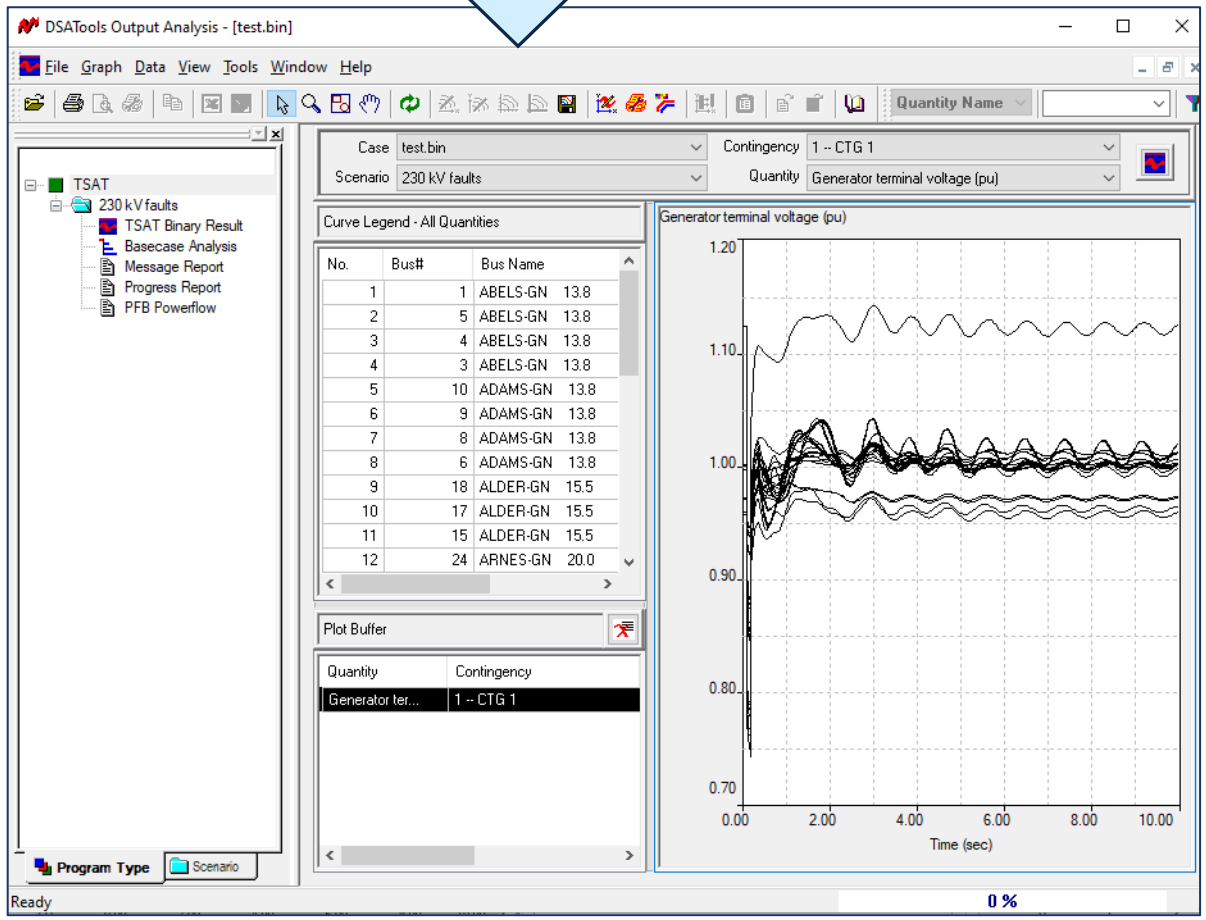
- The Output Analysis Module (DSAOA) can be used to manage, view, plot, and analyze the results of VSAT & TSAT
  - Different tabular display and plotting options
  - Highly customizable displays
  - Data/graph import/export options
  - Analysis tools
  - Script to generate plots in batch mode

(a similar application, OAM, is included with SSAT)

OUTPUT ANALYSIS – DSAOA (CONT'D)

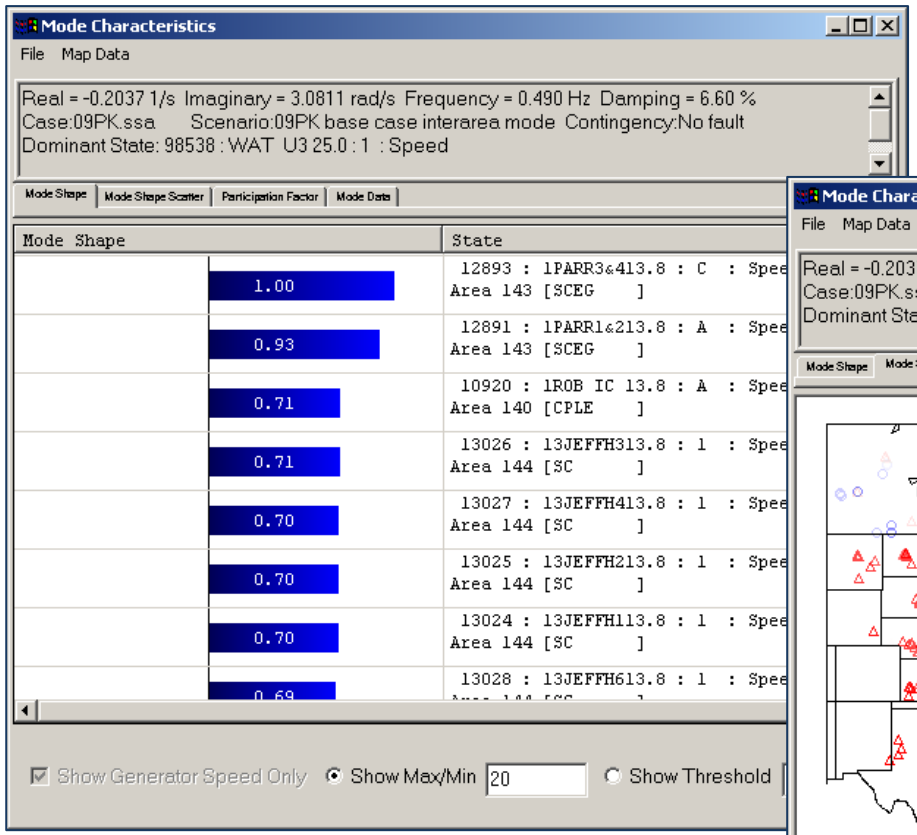


- VSAT P-V curves
- TSAT simulation results

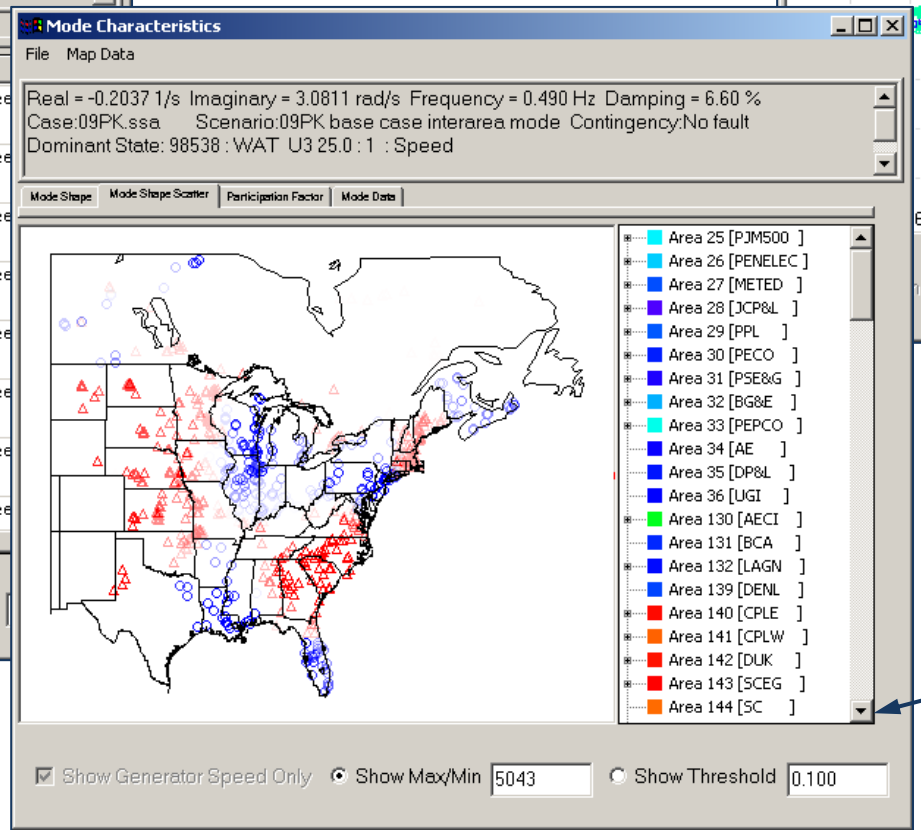


OUTPUT ANALYSIS – OAM

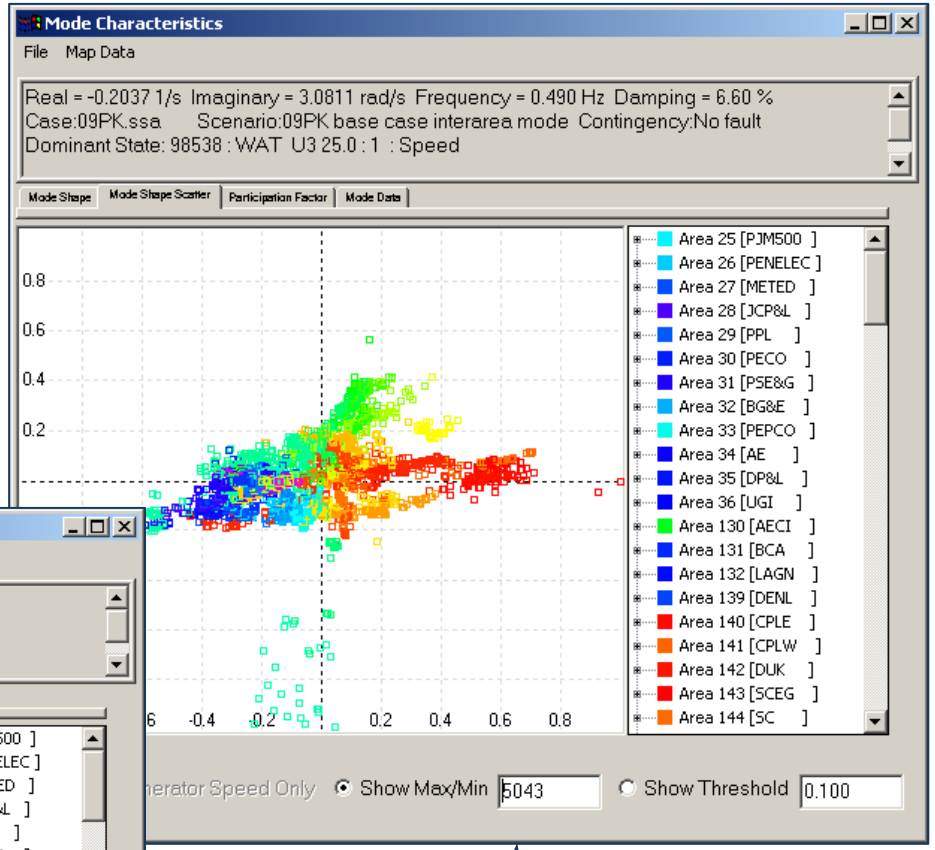
- In SSAT, mode shape can be viewed in three different ways:



- Bar Chart



- Geographical

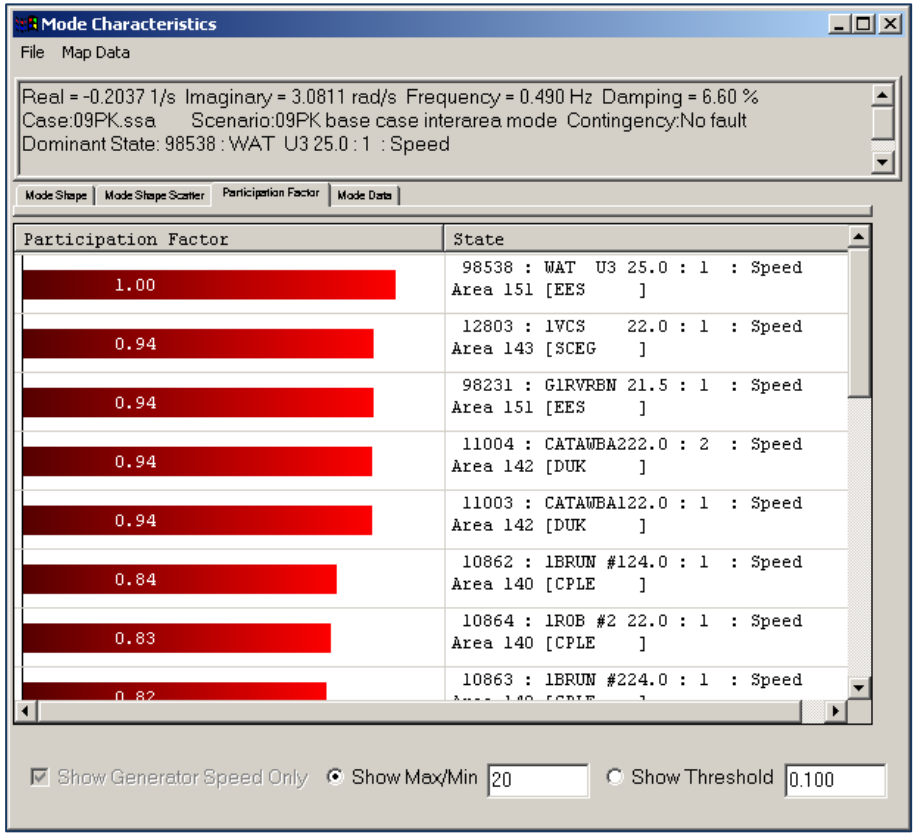


- Scatter

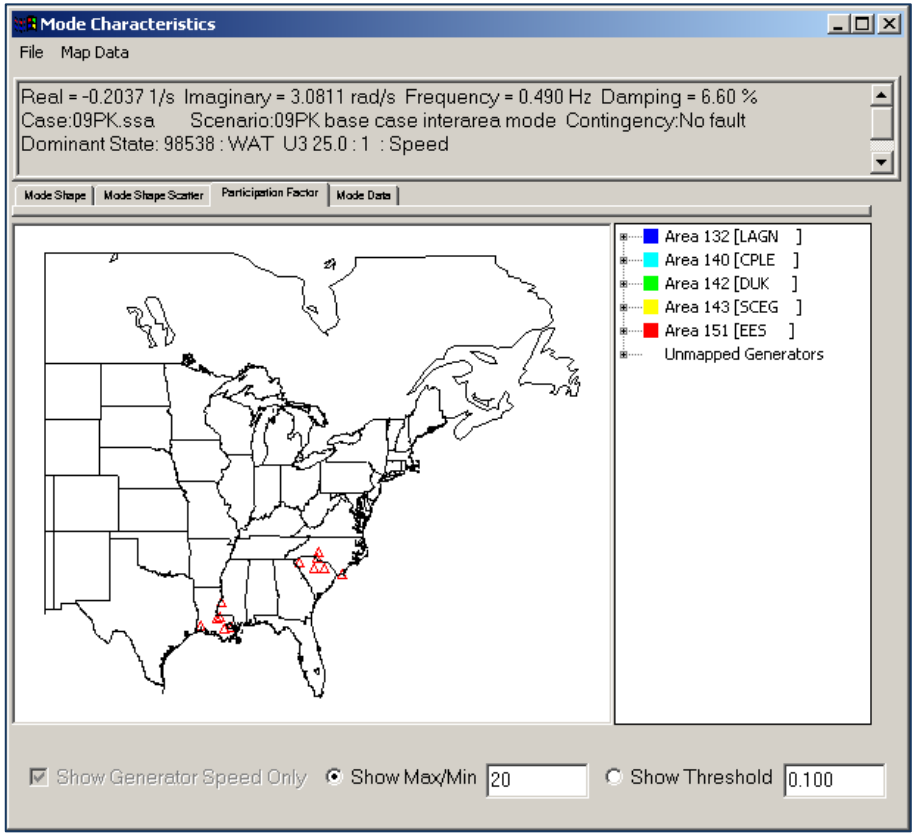
Customization options available

OUTPUT ANALYSIS – OAM (CONT'D)

- In SSAT, participation factors can be viewed in two different ways:



- Bar Chart



- Geographical

ON-LINE DYNAMIC SECURITY ASSESSMENT (DSA) OVERVIEW

WHAT IS ON-LINE DSA?

- **Objective:** to ensure secure operation of a power system
- What's the **difference** from the conventional (off-line) approach?
  - Starts from a **real-time captured** (or forecasted) system condition
  - Considers all applicable security criteria and credible contingencies ("**compliancy**")
  - Include high fidelity system models ("**model-based analysis")**
  - Perform analyses **automatically**, **periodically**, and **efficiently**
  - Present results in simple, straightforward way – security margin, operational limits, remedial actions required, etc

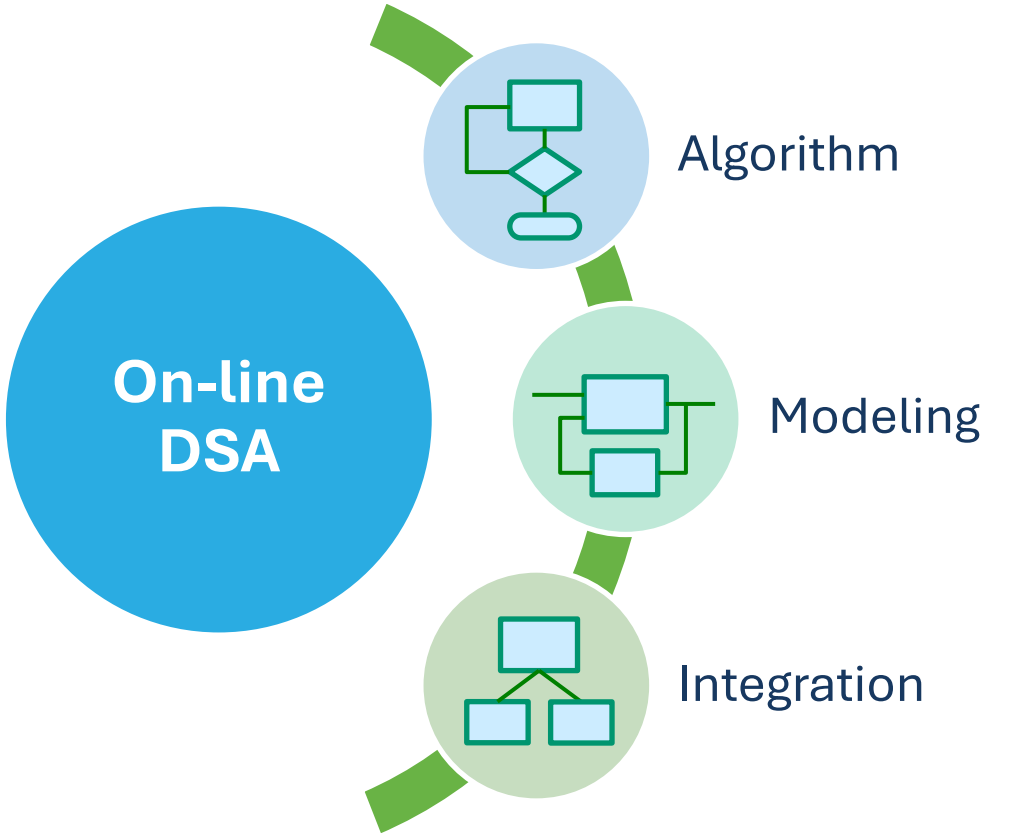
Powertech's on-line DSA software is **proven choice for online DSA** running in many control rooms around the world

WHY DO DSA ON-LINE?

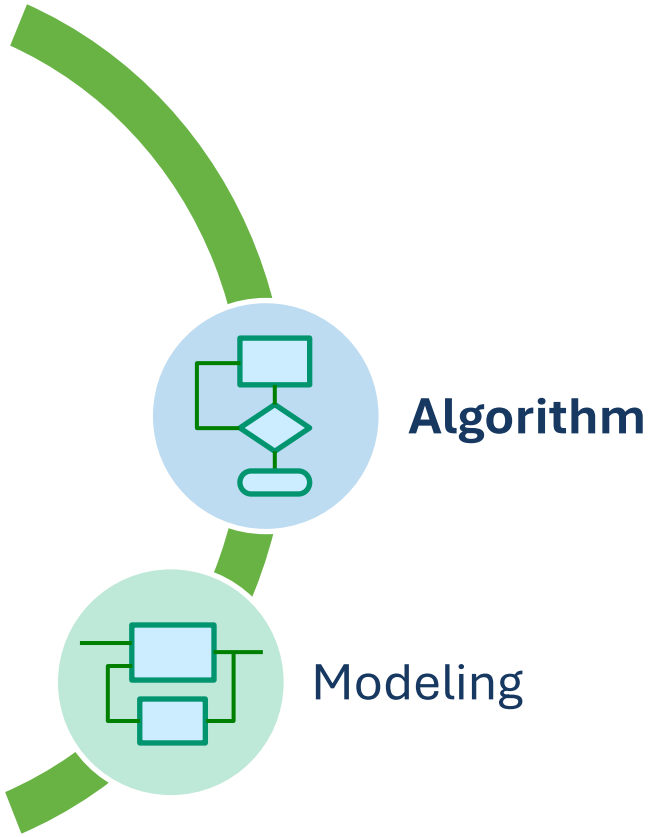
- Improved grid security . . . **prevent blackout!**
- **Better situational awareness** in control rooms
- Dealing with unpredictability and uncertainty in grid operation, especially with **high renewable penetration**
- Maximized use of transmission capacities
- Dealing with congestion in power market
- Minimized generation and reserve cost
- Reliability compliance
- Fast and accurate controls
- *More . . .*



THREE KEY ELEMENTS IN ON-LINE DSA



ALGORITHM



Static analysis

- **P-V**
- **Q-V**
- **Continuation powerflow**
- **Modal analysis**
- OPF

Dynamic analysis

- **Time-domain simulations**
- Direct methods (EEAC, BCU, etc)
- **Eigenvalue analysis**
- Trajectory sensitivities

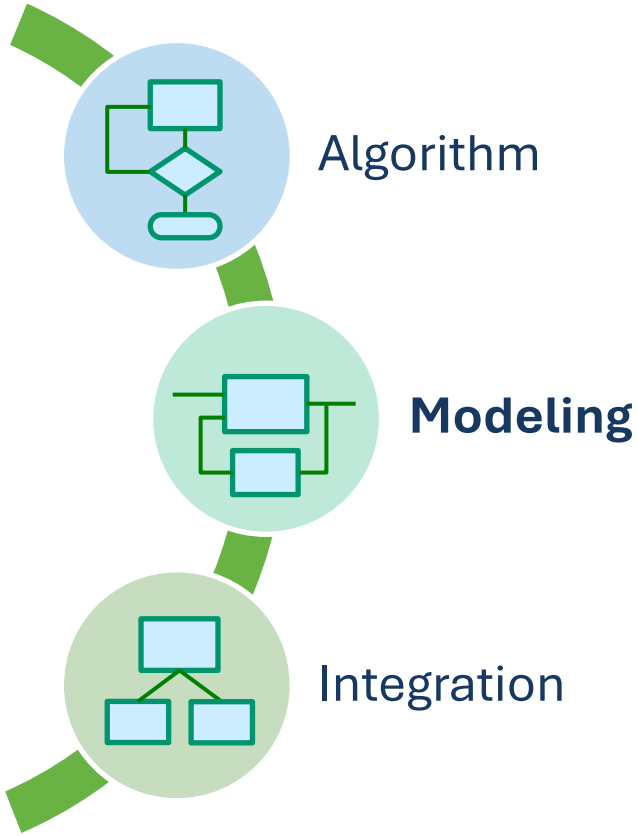
Special analysis

- **Transfer limit analysis**
- **Remedial controls**
- **Distributed/parallel /** cloud computations
- **Real-time closed loop controls**
- Big data analysis
- Probabilistic analysis
- AI methods

Blue – Implemented in Powertech DSATools

New methods are being proposed and applied continuously

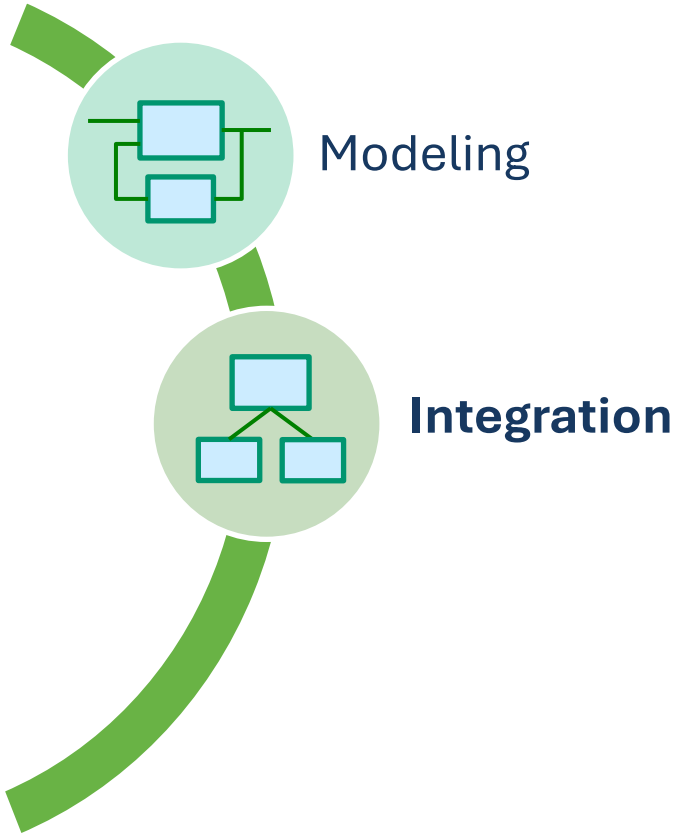
MODELING



In addition to the models considered in the conventional off-line DSA, additional models or modeling techniques are required for on-line DSA

- **Examples:**
  - Models need to handle various operating points
  - Node/breaker modeling
  - RAS and SPS models
  - Realistic contingency sequences
  - Real-time equipment status
  - Dynamic rating
  - Renewables
  - External system equivalencing

INTEGRATION



Integrated Control Room Application

- Real-time
- Forecast
- Look-ahead
- Market
- PMU/WAMS
- Training
- Testing
- Study
- Redundant/Backup
- Result depository

On-line DSA

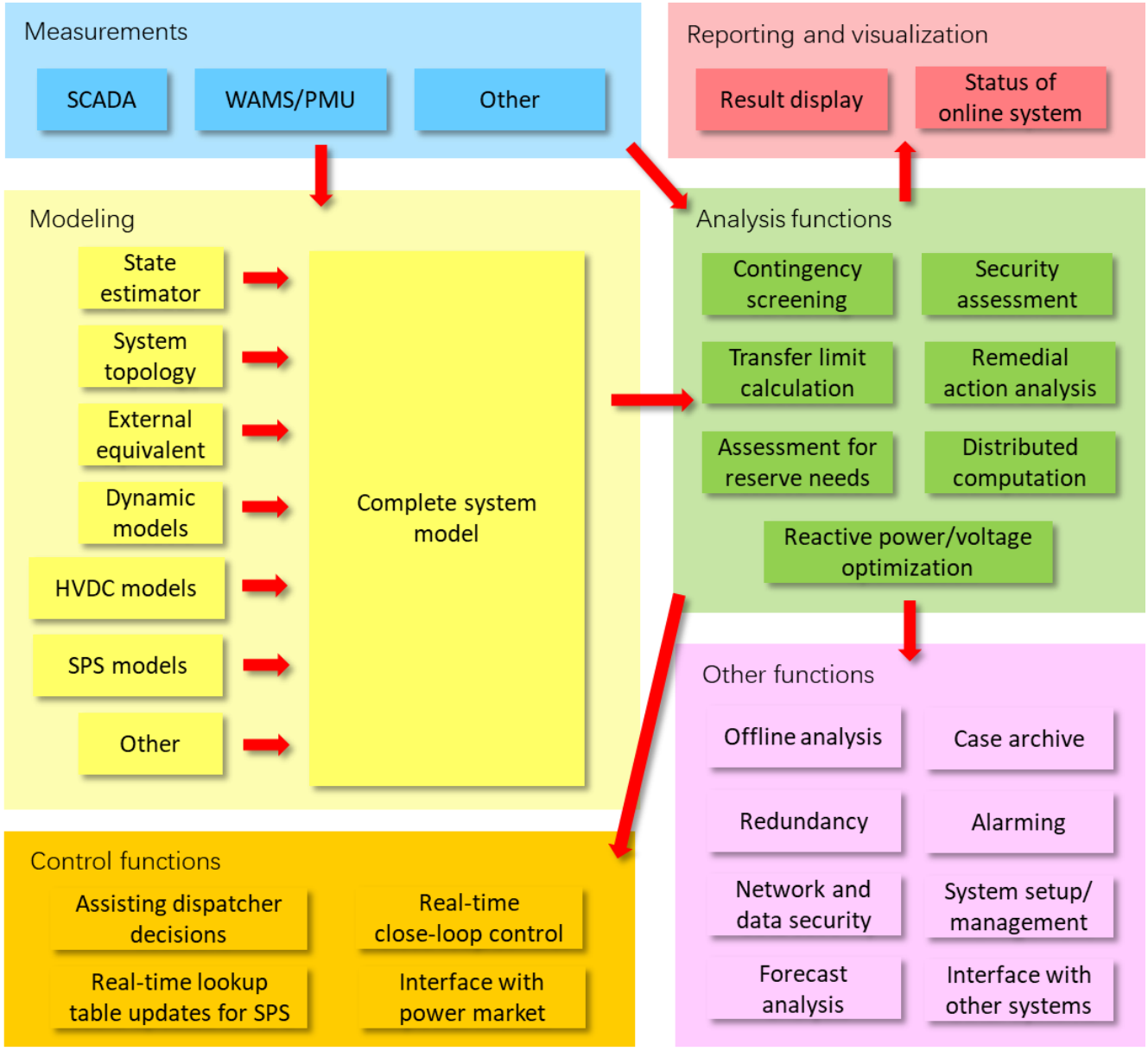
- RT model recondition/ match
- Auxiliary data management
- Resource and data sharing
- Analysis scenario creation
- Performance monitoring
- Result processing and display
- Failover and security

Auxiliary Databases

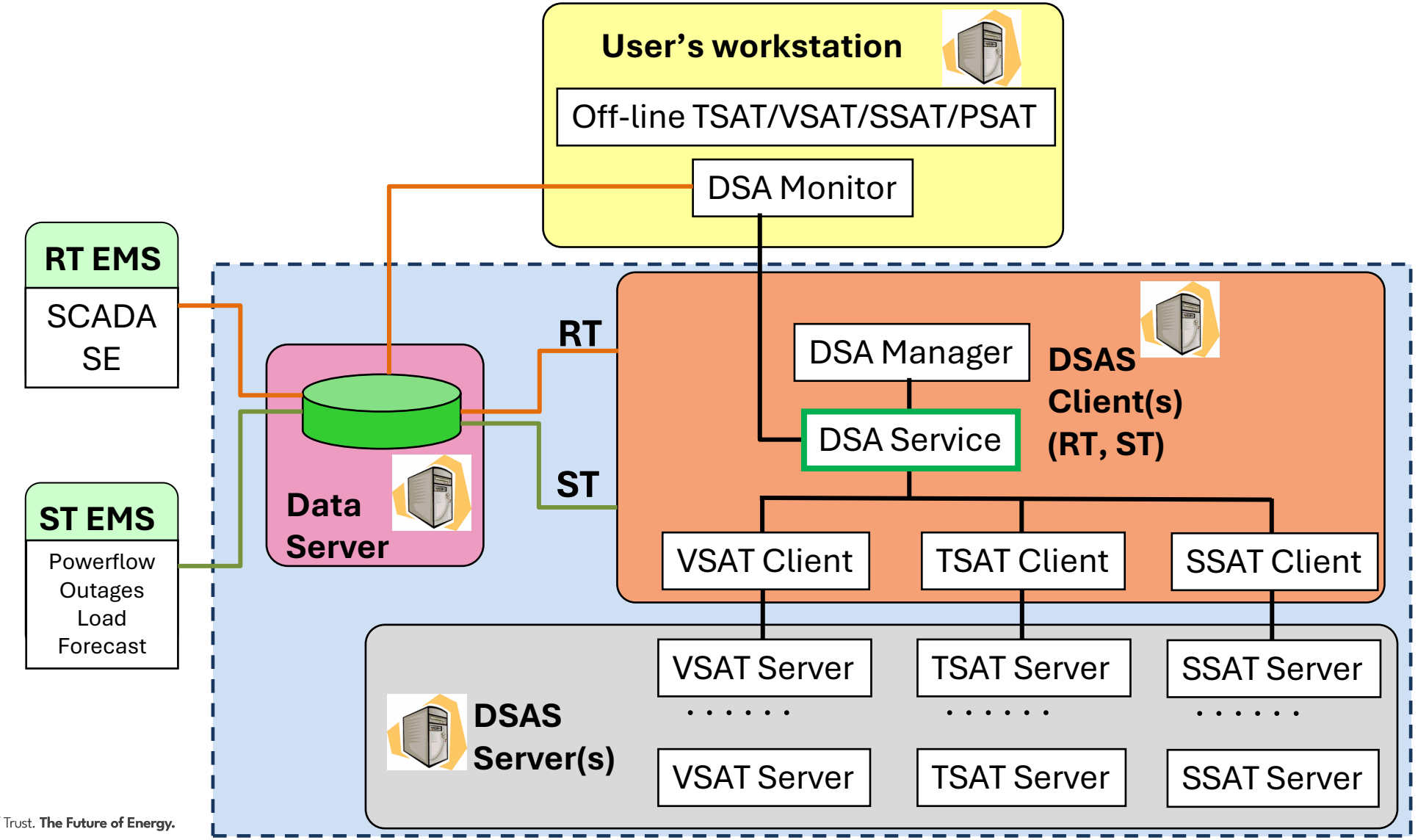
Computation Engines

- Distributed/parallel / cloud computing

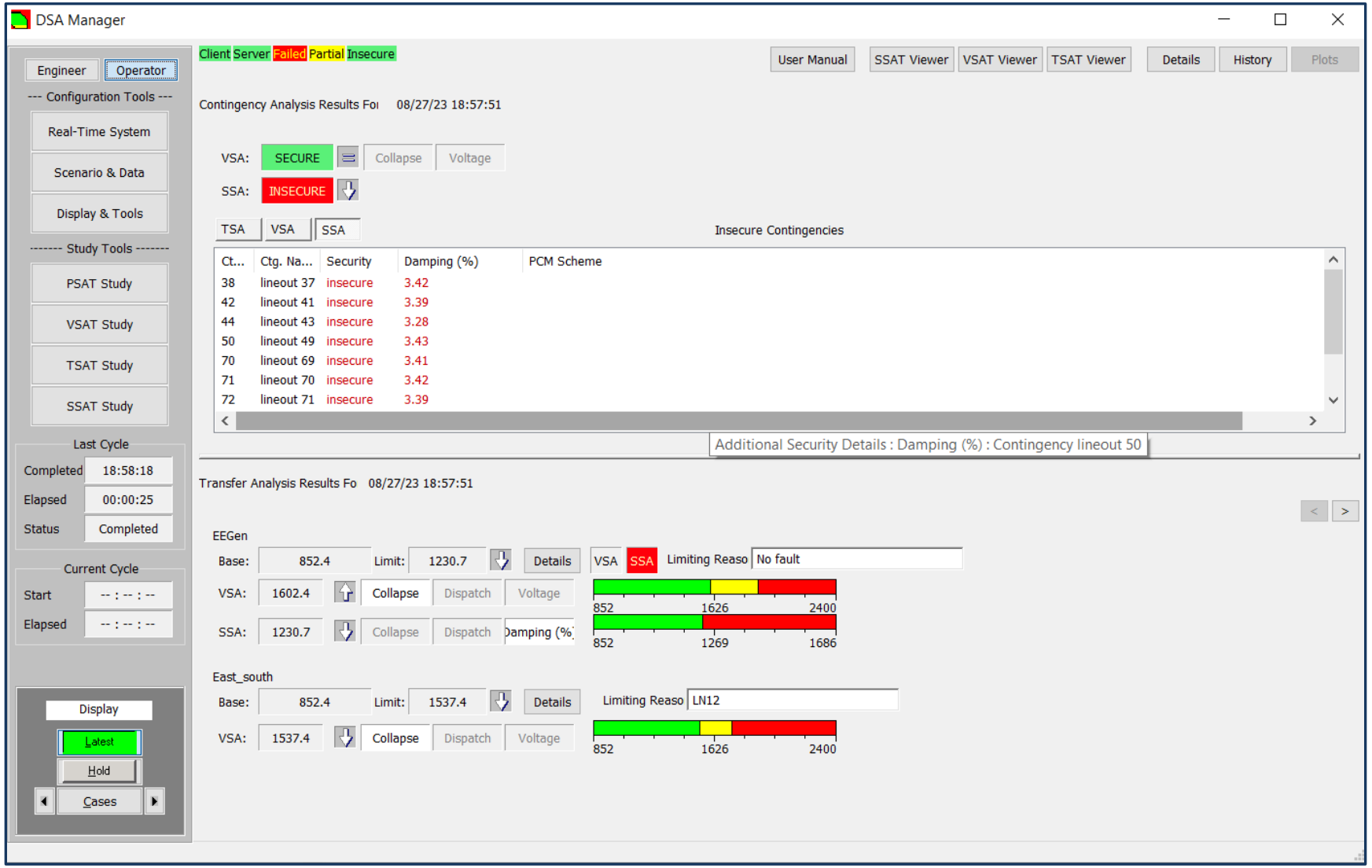
ON-LINE DSA FUNCTIONAL MODULES



ON-LINE DSA - SOFTWARE / HARDWARE OVERVIEW



DSA MANAGER USER INTERFACE



BENEFITS



Reliability

- On-line DSA has been used as a **primary tool** to ensure the secure operation of power grids



Environmental

- On-line DSA can help increase the **level of renewable generation** that a system can take from security perspective



Economical

- On-line DSA gives **more accurate (and usually higher)** power transfer limits, thus helping **reduce the cost** in system operation

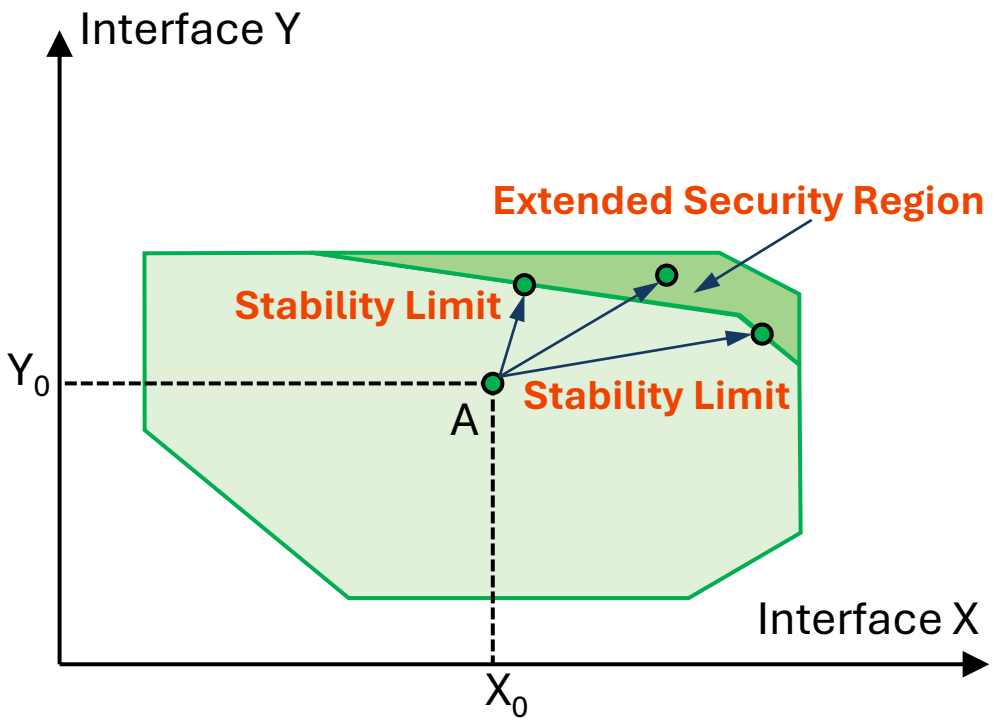
ON-LINE DSA APPLICATION EXAMPLES

EXAMPLE USE CASES & APPLICATIONS

- System security status and margin monitoring
- Security limit determination
- Recommendations for preventive/corrective control actions
- Impact assessment of renewables on system security
- Verification of system protection schemes (SPS)
- Transaction settlement in power market
- Determination of active and reactive power reserve
- Scheduling of equipment maintenance
- Support to PMU/WAMS applications
- Calibration and validation of power system models
- Preparation of models for system studies
- Post-mortem analysis of incidents
- System restoration

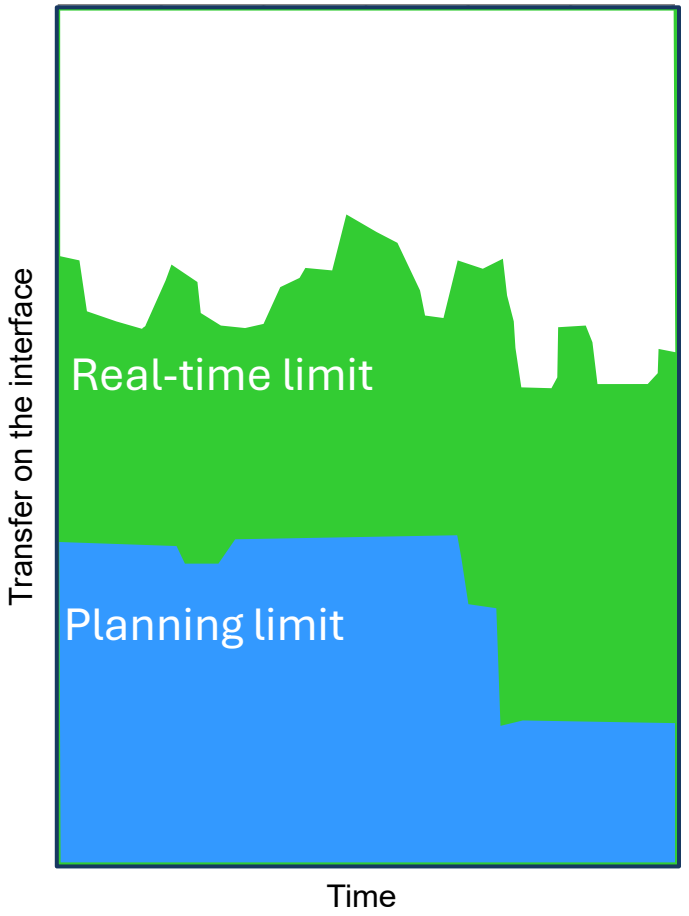
POWER TRANSFER LIMIT ANALYSIS (SECURITY LIMIT)

- Assume that a system is operating at a condition (“A”)
  - Measured by some system parameters (e.g. interface flows)
  - Determined to be secure for all credible contingencies
- We need to know what is the secure region within which the system can move
  - Or which direction to move to achieve maximum transfer (e.g. export/import)
- If the system needs to move to an insecure point for some reason, what needs to be done to ensure the system security
  - Determined, for example, by adjusting special protection system settings



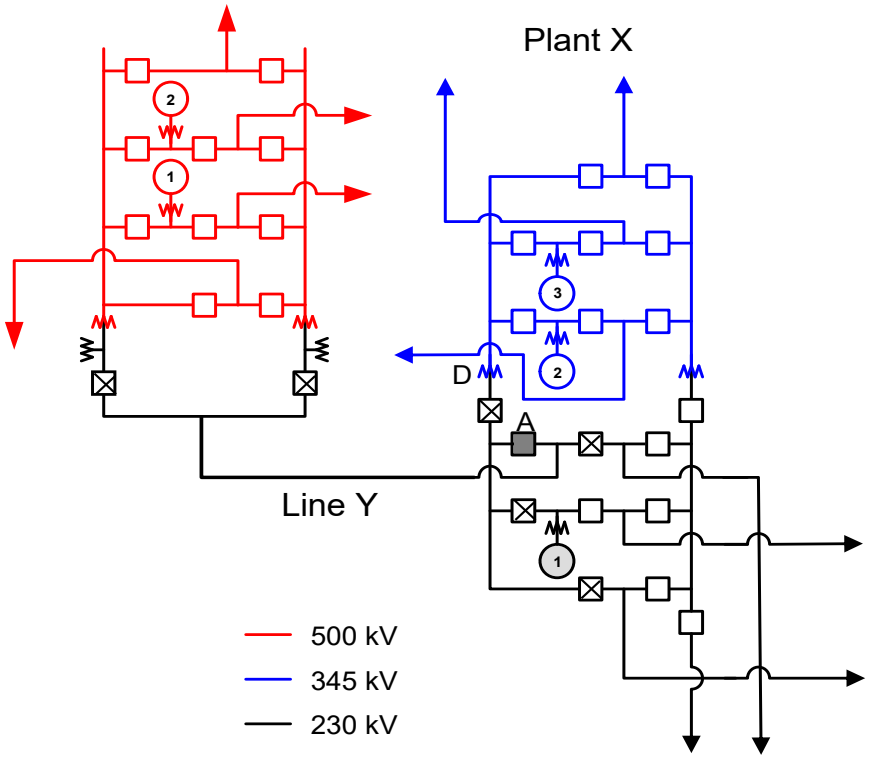
BENEFITS OF SUCH ANALYSIS

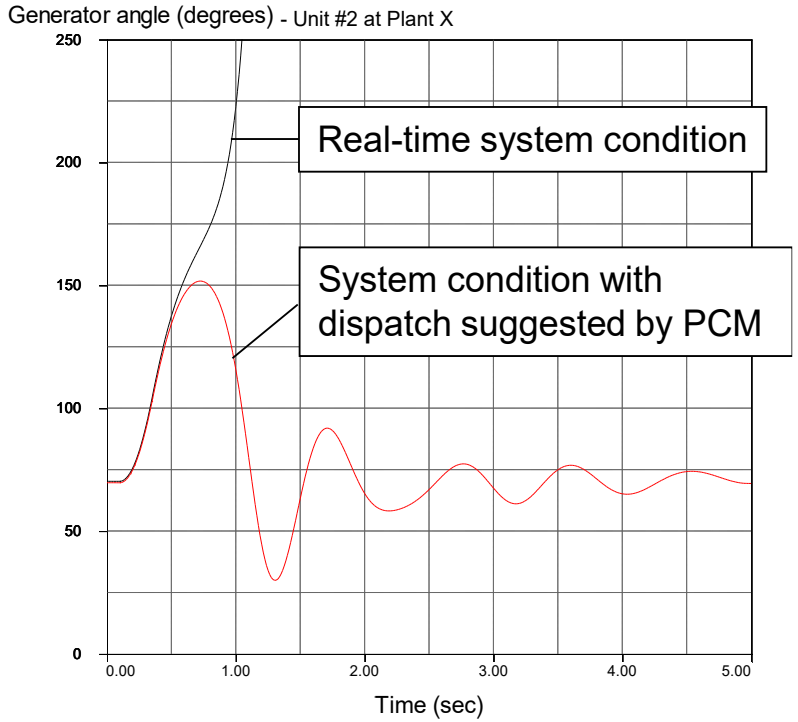
- Increased degree of system security
- More opportunities for power market participants
  - Real-time calculated limits are generally higher than the planning limits
- Lower cost for congestion management
  - The monthly credit paid by IESO (the ISO in Ontario, Canada) averaged $11.7m during 2003-2007
- Higher level of renewable export
  - A study by ERCOT (the ISO in Texas, USA) discovered that the limits based on-line DSA could allow as much as 400 MW more wind export from its western region (more than 10% of the total transfer capacity)



RECOMMEND REMEDIAL CONTROL ACTIONS

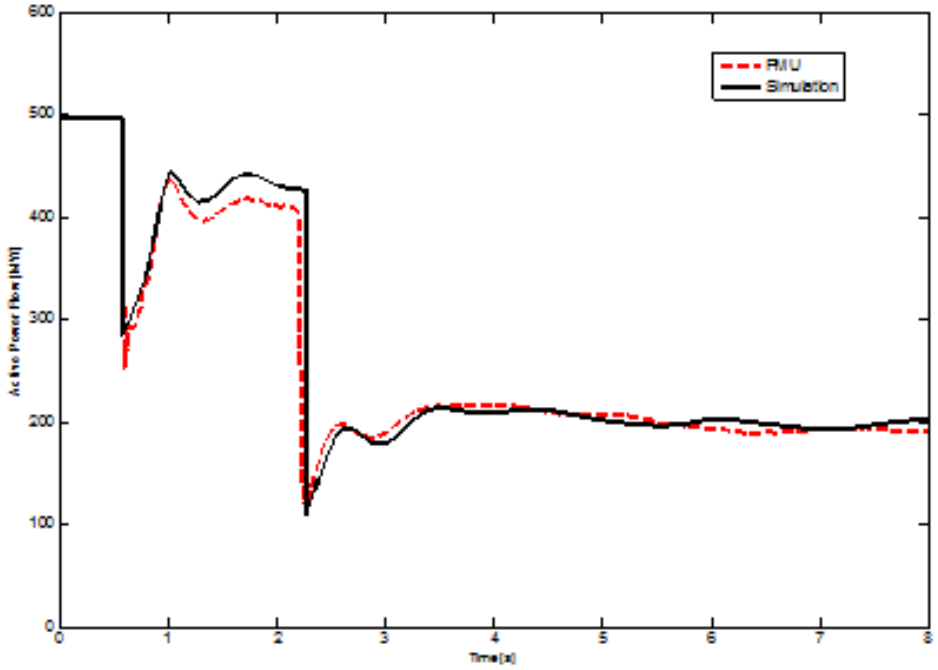
- A case obtained from an actual on-line DSA system which includes a Preventive Control Measure (PCM) module
- The PCM module was activated to investigate a transiently unstable contingency





MODEL VALIDATION

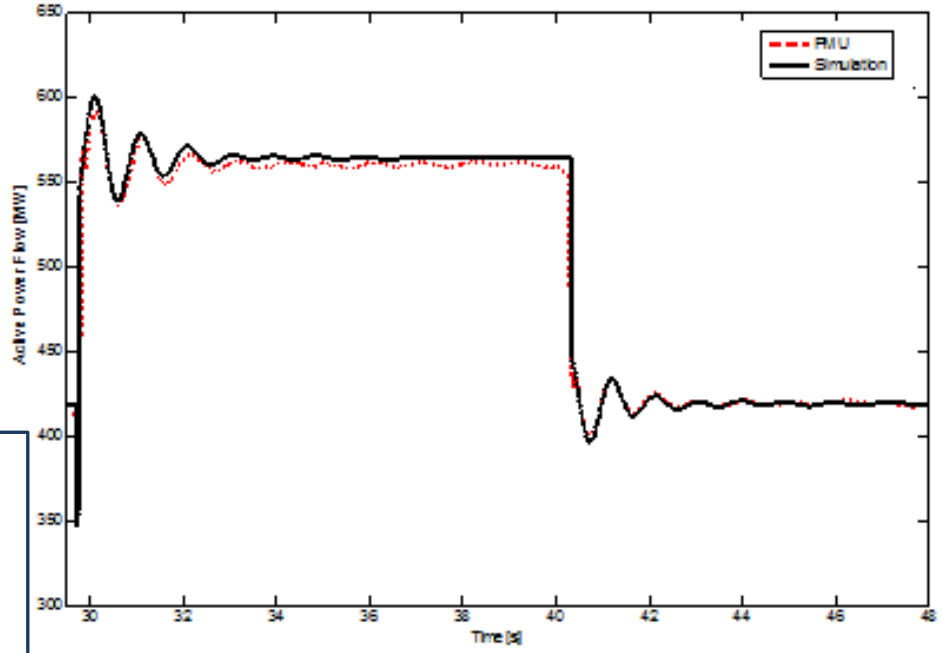
- Two model validation examples done by ISO New England
  - Fault at a 345 kV line followed by successful reclosing
  - Blocking of one pole of an HVDC link due to a fault followed by the blocking of the second pole



MW flow on an AC line

PMU recording

On-line DSA simulation



APPLICATION EXAMPLE: NATIONAL GRID UK

- Motivation for on-line stability analysis (OSA)
  - 60 GW island system with DC Interconnections to Europe
  - Renewables brought more uncertainty
  - **Off-line studies no longer adequate**
- Capabilities of the OSA
  - Including both transient stability and damping criteria
  - 3,000 bus system model
  - 2,000 contingencies (N-1, N-2, detailed)
  - 15-min computation cycle with 200 CPU cores

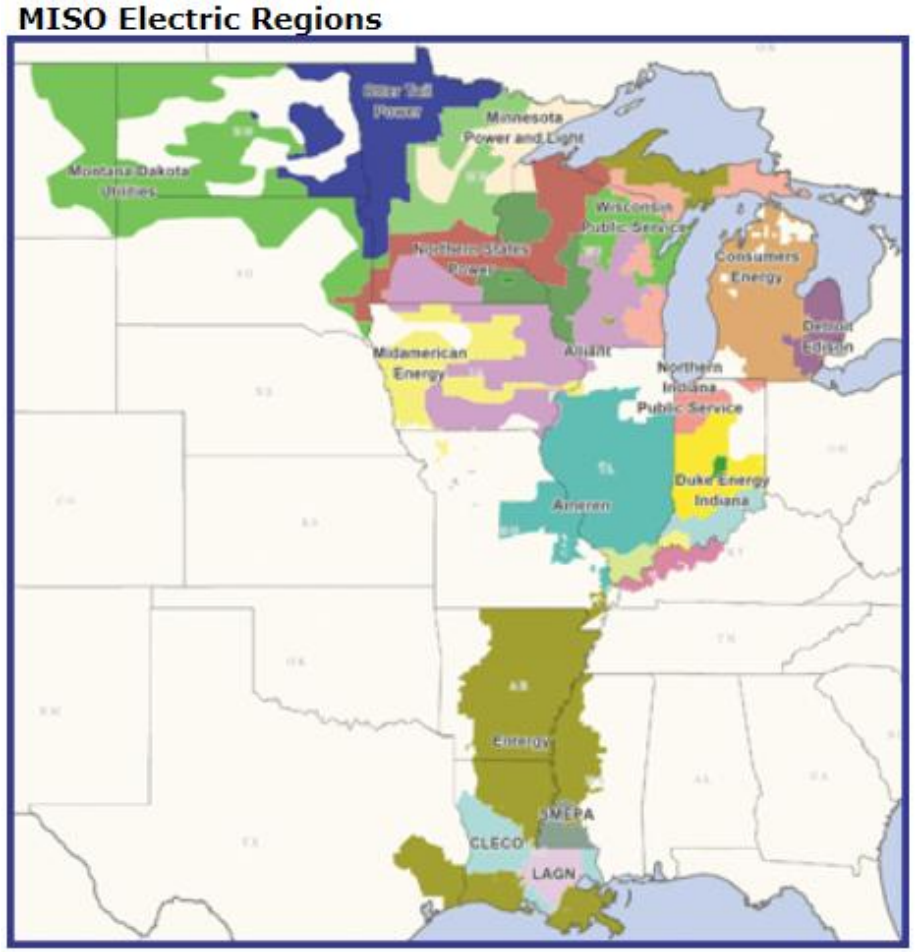


APPLICATION EXAMPLE: NATIONAL GRID UK (CONT'D)

- A special **study mode** enables engineers to quickly perform studies for various scenarios based on real-time system conditions
- Benefits achieved
  - **Transient stability risks** identified, which were not previously known
  - **Damping risks identified** including one involving a group of 8 GW generation, which were not previously known
  - Instability risks at special operation conditions (hydro units in pumping mode, circuit switching for voltage control, etc.)
- Development work in 2016~2017 focused on new HVDC modeling and moving the tool to off-line study
  - 5 new HVDC interconnectors (and counting)
- Online VSA added in 2022-2023

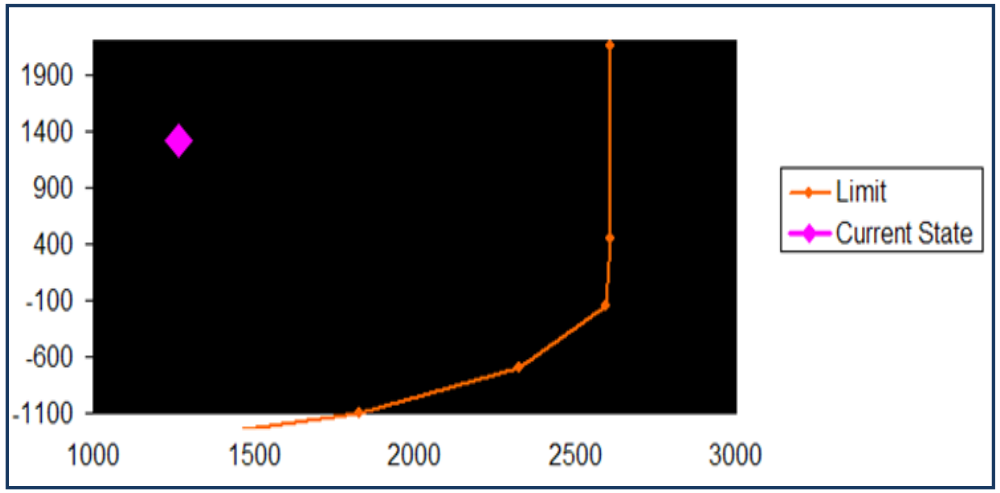
APPLICATION EXAMPLE: MID-CONTINENT ISO (MISO)

- MISO is an Independent System Operator (ISO) in Mid-West US
- Installed capacity: 190 GW (including 16 GW of wind)
- Peak load: 130 GW
- 8 HVDC links (3 back-to-back)
- Eastern region of MISO experiences mostly **voltage stability** problems while **transient stability** is the main concern in its western region
- Real-time network applications are performed for very large real-time system models (43,000+ powerflow buses)
  - SE solves every ~ 90 sec using 293,000+ real-time measurements

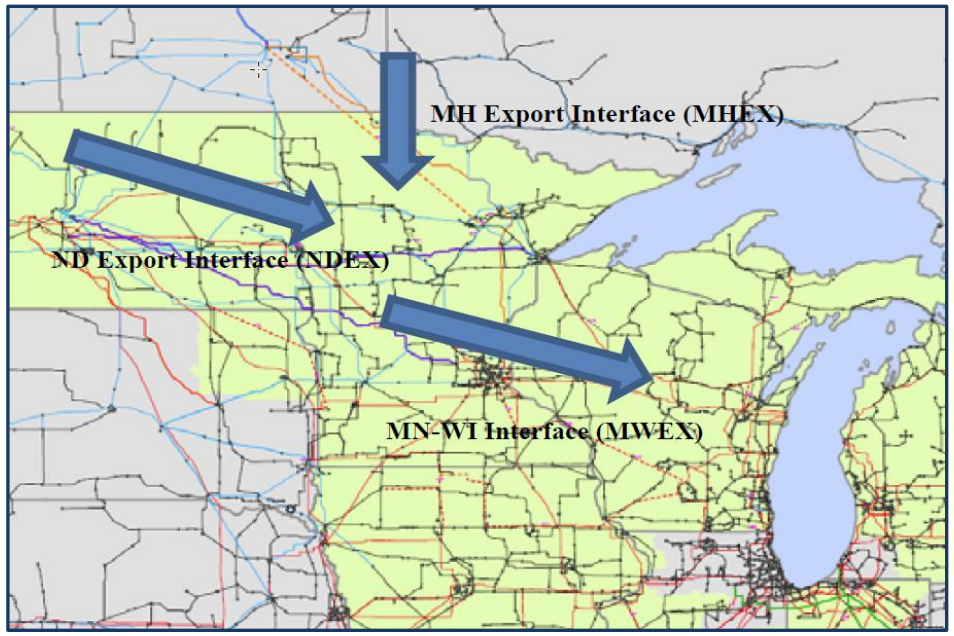


APPLICATION EXAMPLE: MISO (CONT'D)

- On-line VSAT has been in service since 2003; On-line TSAT was in production mode May 2012
- The main objective is to monitor and calculate real time voltage and transient stability limits on the key interfaces
- For on-line TSAT, the **archived** real-time cases are also being used for MISO operational planning studies



2-D voltage stability limit nomogram



APPLICATION EXAMPLE: MISO (CONT'D)

- Ice storm in winter 2005
  - MISO system was hit with an ice storm in winter 2005
  - A total of **53** transmission lines 115 kV and higher were tripped at different stages of this event
  - On-line VSA was a critical tool used during this event to manage the system operation, as a result, **no forced load shedding was performed**
  - MISO recognized that this was impossible without on-line VSA
- **Economic benefit (one of the key stability limited interface)**

Year	Interface Limit Calculated	# Real-Time Binding Hours	Total Real-Time Congestion Costs (Million $)
2014	Offline	502	31
2015	Offline	7	0.7
2016	Offline	3	0.1

APPLICATION EXAMPLE: EIRGRID

- The Irish national grid company
- Small system but with high wind power penetration
  - Can handle up to 75% variable renewable energy (predominantly wind power) on the grid at any time
  - Wind penetration often exceeds 70%
- The question: what is the highest amount of wind generation allowed at any given instant of time?
  - Subject to thermal, voltage, frequency, and stability criteria

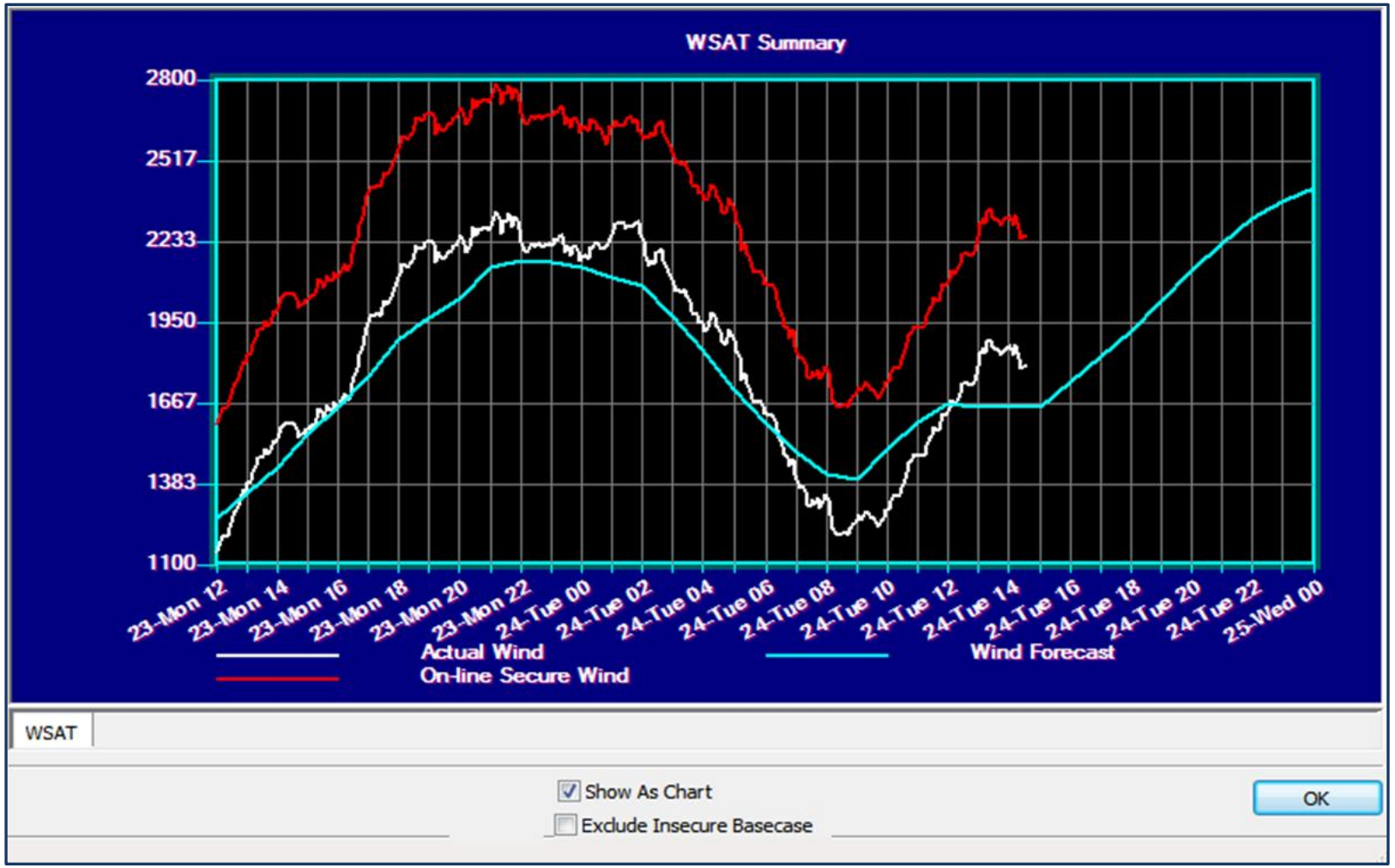


APPLICATION EXAMPLE: EIRGRID (CONT'D)

- WSAT was developed at EirGrid control center as a real-time application
  - Based on Powertech's VSAT and TSAT
- Provides max allowable wind generation for
  - Real-time condition
  - Forecast condition (integrated with wind forecast data)
- Special modeling considerations
  - Wind modeling
  - HVDC Link modeling (to Scotland)
  - VSC DC Link modeling (to Wales)

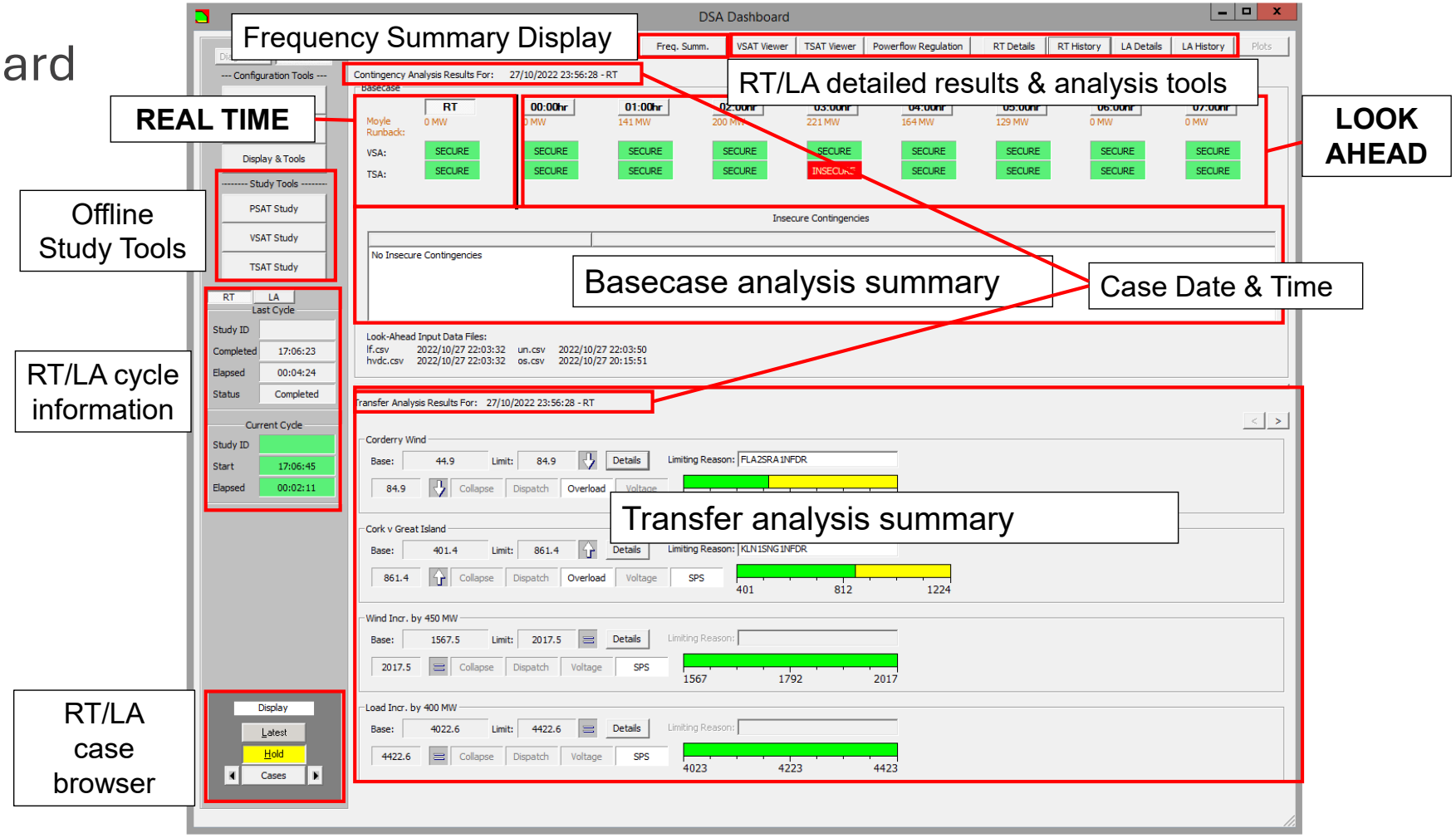
APPLICATION EXAMPLE: EIRGRID (CONT'D)

- WSAT secure wind graph



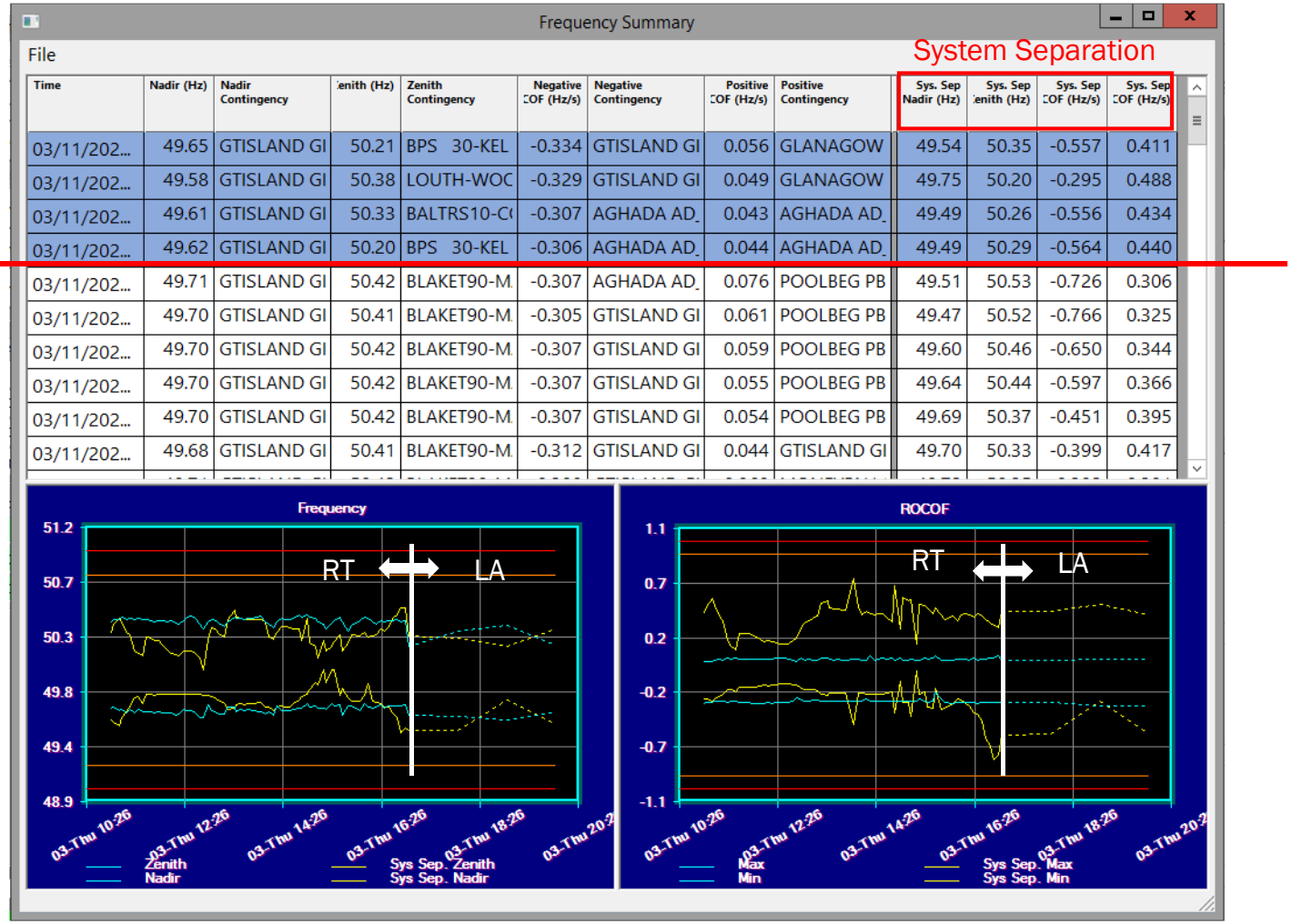
APPLICATION EXAMPLE: EIRGRID (CONT'D)

- Look-Ahead Dashboard



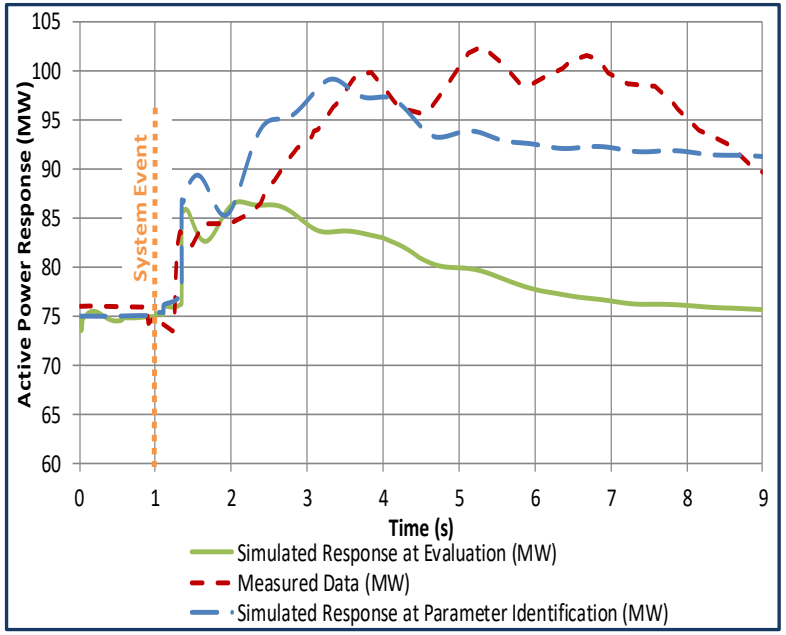
APPLICATION EXAMPLE: EIRGRID (CONT'D)

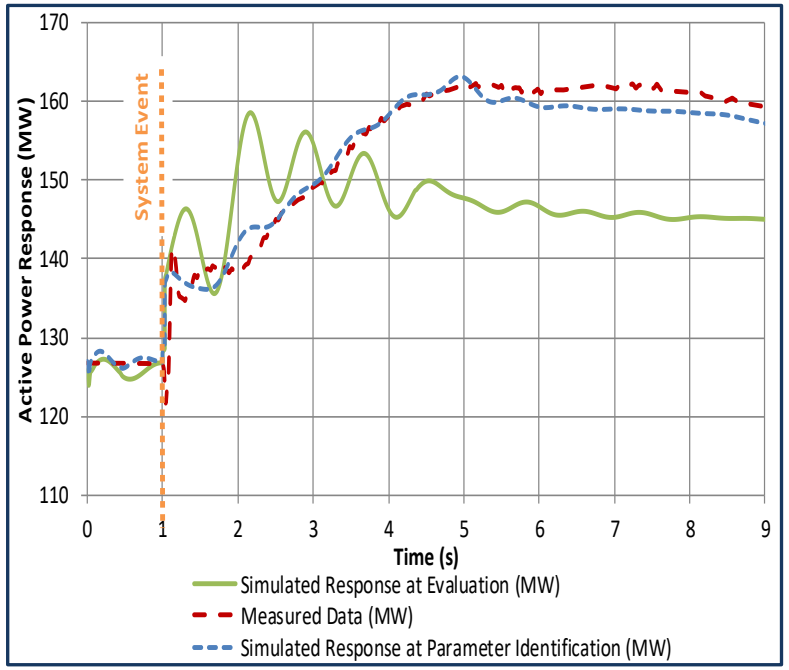
- Look-Ahead Frequency Summary

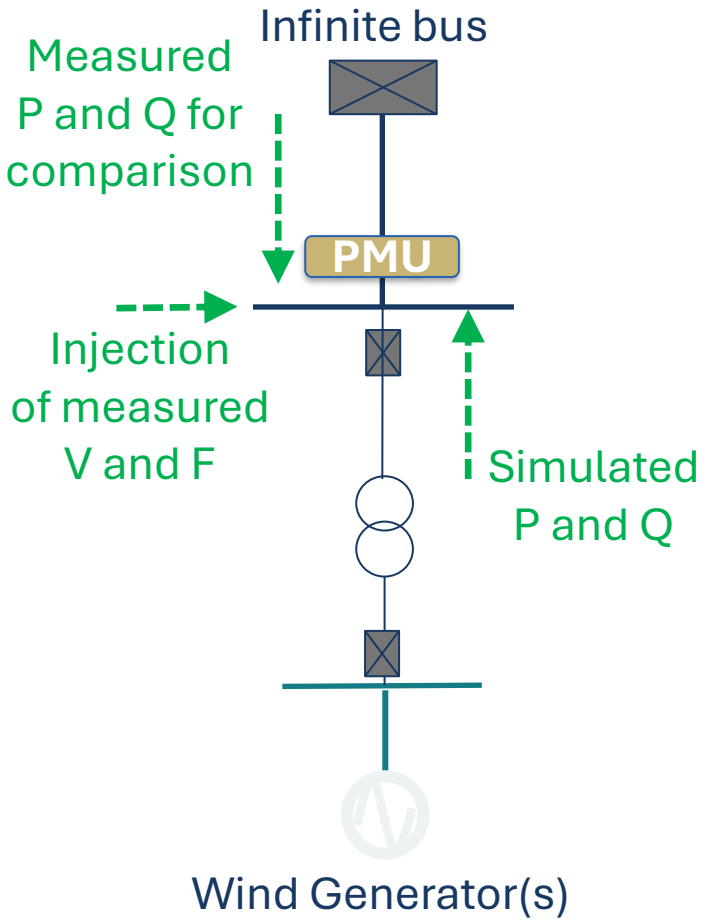


APPLICATION EXAMPLE: EIRGRID (CONT'D)

- Wind Model Validation

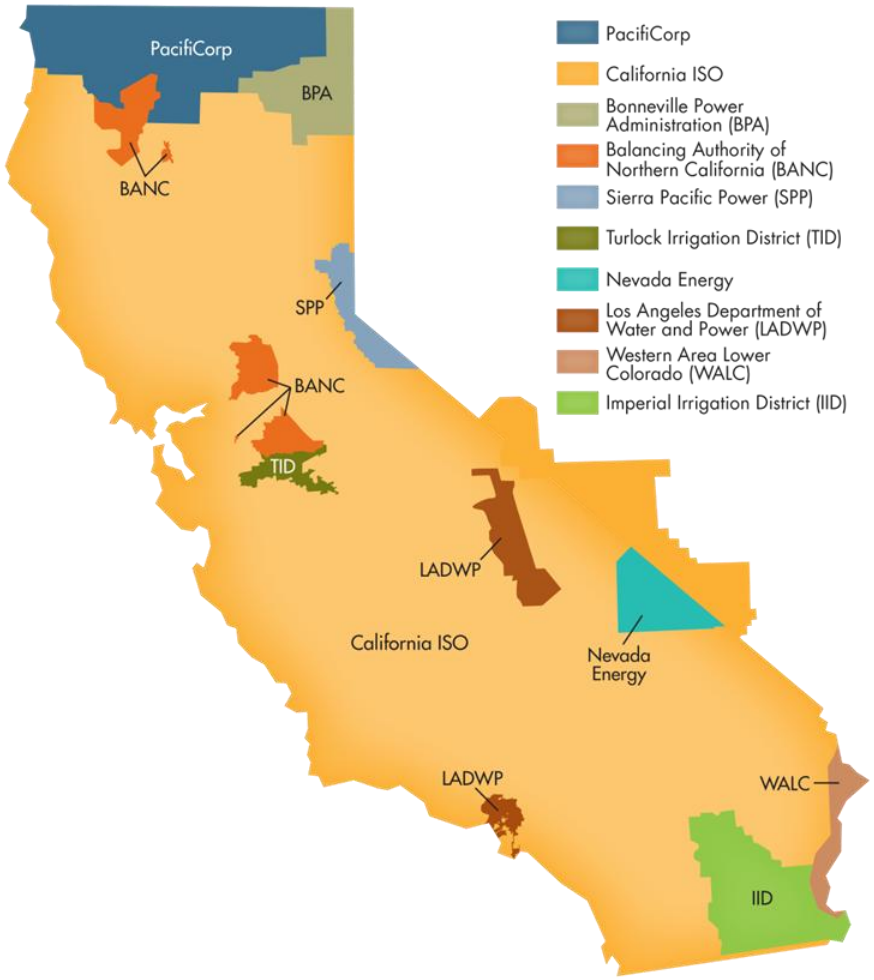






APPLICATION EXAMPLE: CALIFORNIA ISO (CAISO)

- Manages approximately 80% of California's electricity load
- Installed capacity: ≈61 GW
- Peak load: ≈ 50 GW
- **27,000+** market transactions per day
- 26,024 circuit-miles of transmission lines
- 30 million people served
- 307 million megawatts of electricity delivered annually



APPLICATION EXAMPLE: CAISO (CONT'D)

- EMS model covers all WECC region
- Detail model for California and neighboring entities
- Simplified model for the rest of WECC. Full loop is explicitly modeled for 200 kV system and above
- Model size

	CAISO Real-Time Model	WECC Summer 2011 Case
Bus	7300	17000
Load	2900	8525
Generator	2200	3433

- Significantly increased in the past couple years
- Inter-mountain and PDCI HVDC links and Trans-bay VSCDC link

APPLICATION EXAMPLE: CAISO (CONT'D)

- On-line DSA system based on Powertech's TSAT
- Multiple sub-systems:
  - Real-Time mode
  - Look-Ahead mode
  - Short-Term Unit Commitment mode
  - Day-Ahead mode
- Real-time mode cycling time: 15 minutes
  - 60 contingencies for basecase analysis
  - 12 1-D transfers. Each has around 10 contingencies
  - 48 TSAT computation engines are set up for this computationally intensive task

APPLICATION EXAMPLE: CAISO (CONT'D)

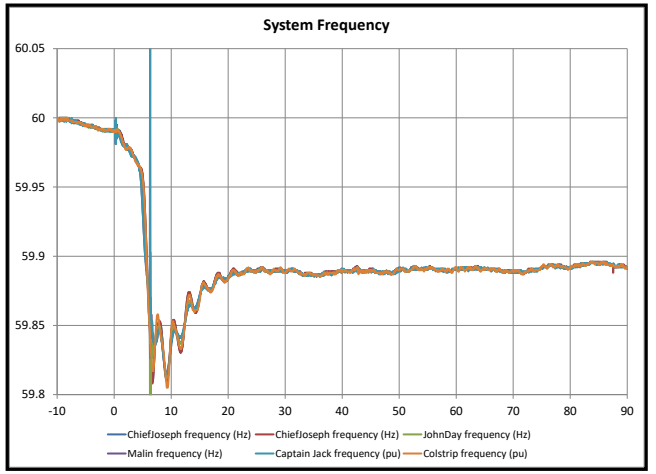
- Special modelling requirements:
  - Wind farms and PV
  - HVDC links and VSCDC link
  - Real-time equipment status
  - SPS
- Operations planning benefits:
  - Very simplified model validation process in a wide area
  - It does not take too much time to setup validation for every event since DSA cases are saved every 15 minutes.
  - With available PMU, model validation is much simpler

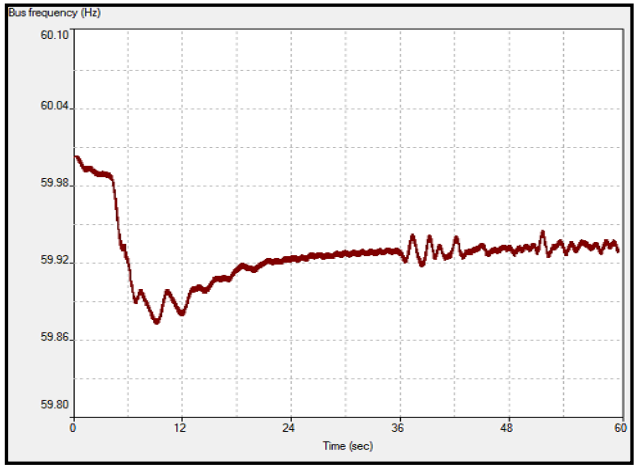
APPLICATION EXAMPLE: CAISO (CONT'D)

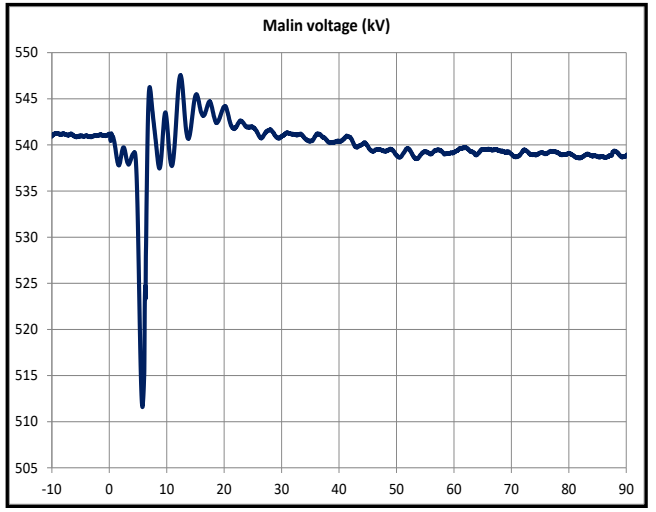
- Real-time benefits:
  - Full situational awareness of the system
  - Knowledge of the current system
  - Knowledge of margin available in the system
  - If modelled properly, SPS modeling allows users to understand the true interactions between SPS
  - No "unknown operating states for more than 30 minute" in Real Time Operations since assessments are continuously performed every 5 minutes (VSA and CA) to 15 minutes (TSA)

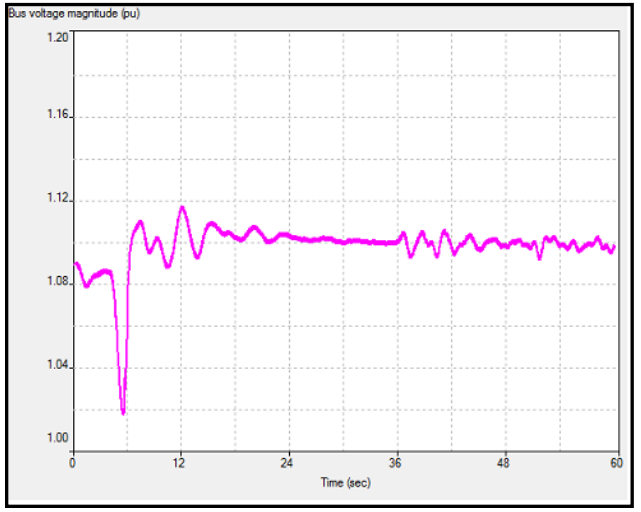
APPLICATION EXAMPLE: CAISO (CONT'D)

- Benchmark Example: July 4, 2012
  – Palo Verde Unit Trip



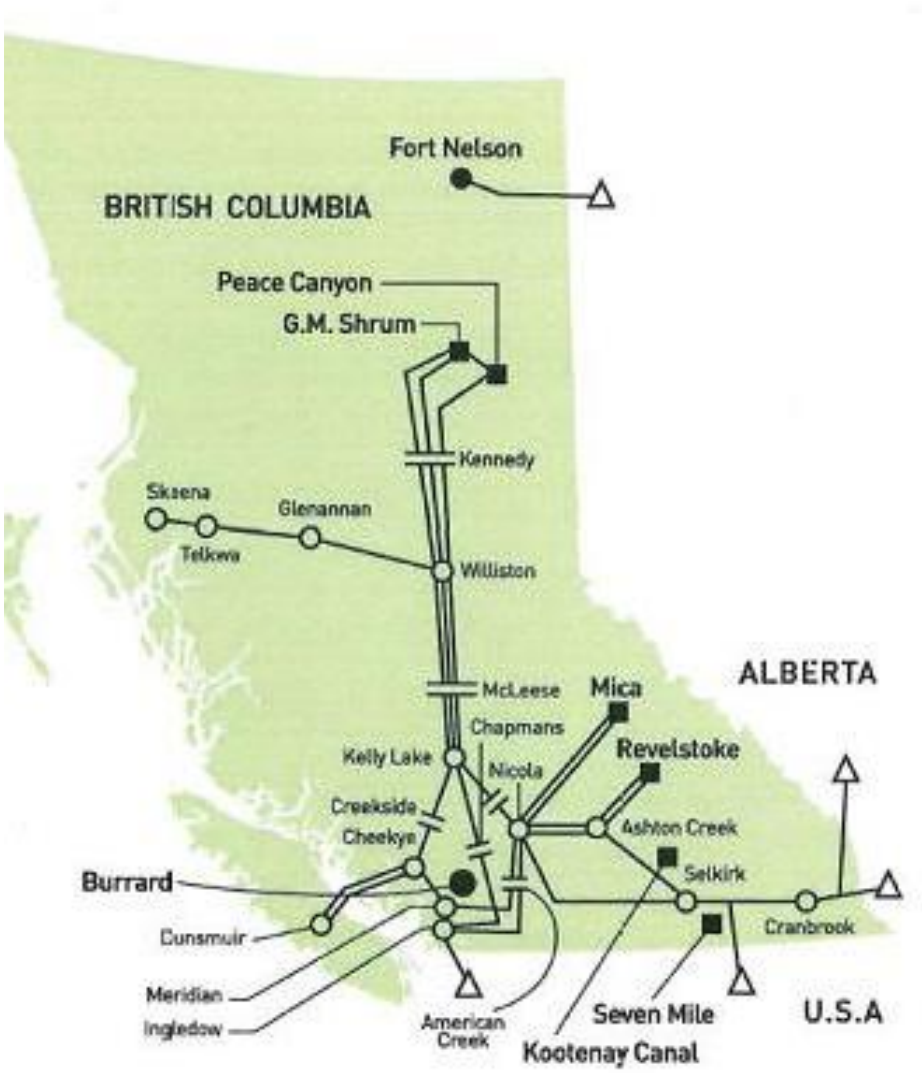






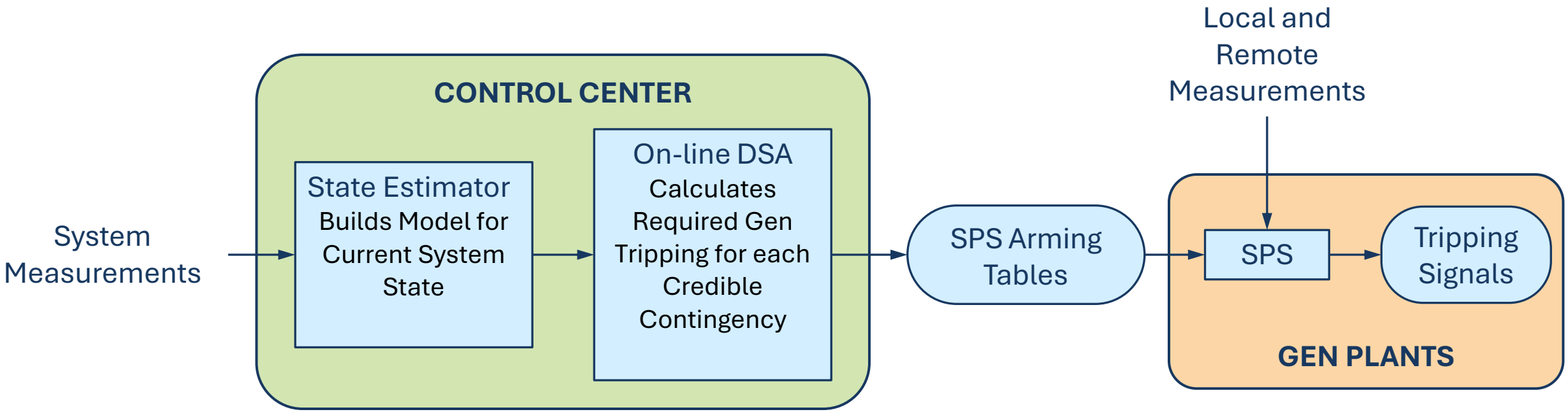
APPLICATION EXAMPLE: BC HYDRO

- Major transfers are constrained by stability
- SPS (generator tripping) is used to ensure system security
- **Real-time generator tripping arming** requires on-line lookup table
- A pattern matching (PM) method has been used which relies on off-line computed lookup tables
- This no longer meets the demand from system operation
- The PM method is being phased out with the on-line DSA technology
- Both voltage and transient criteria are examined



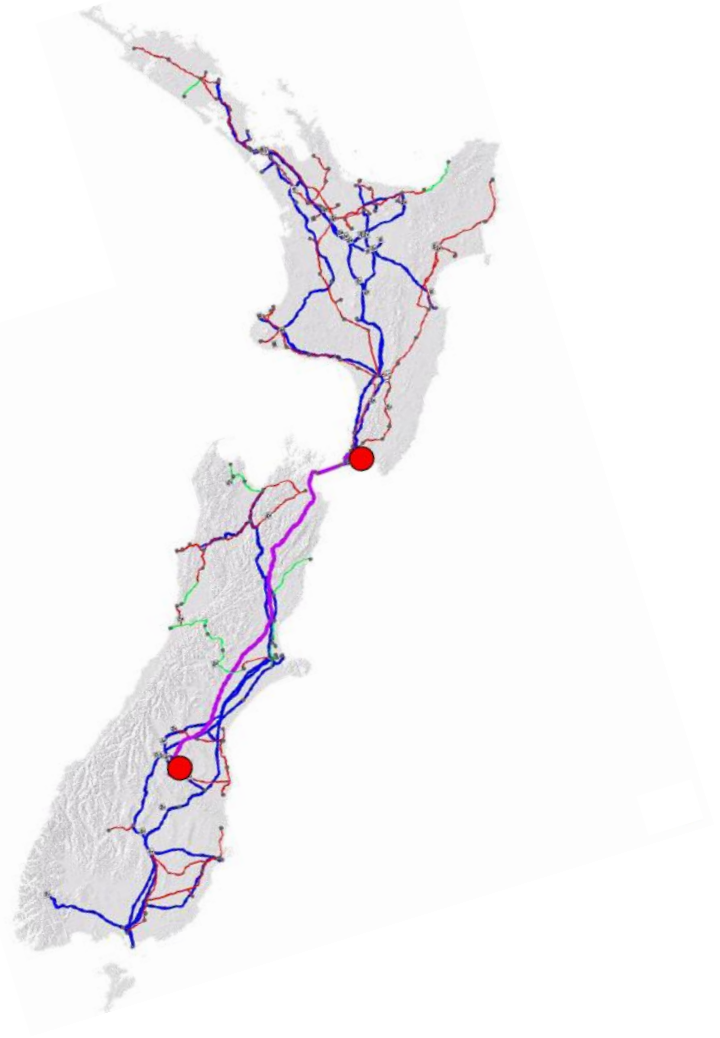
APPLICATION EXAMPLE: BC HYDRO (CONT'D)

- The on-line DSA system runs every 4 minutes to determine
  - Gen tripping required for the current transfer
  - Gen tripping required to achieve max transfer



APPLICATION EXAMPLE: TRANSPOWER NEW ZEALAND

- Ensuring **acceptable frequency response** after the loss of a large unit or HVDC bi-pole is critical to system operation
  - Appropriate **active power reserve** is required
- A **Reserve Management Tool** (RMT) is used for this purpose in NZ Electricity Market
- RMT runs in the market settlement cycle
  - Full time-domain simulations with reduced network model but detailed dynamic models
  - About 30 sec to solve 6 contingencies
  - Tight integration with market data
- Expected cost saving in NZ Electricity Market in 10 years
  - **US$12m**



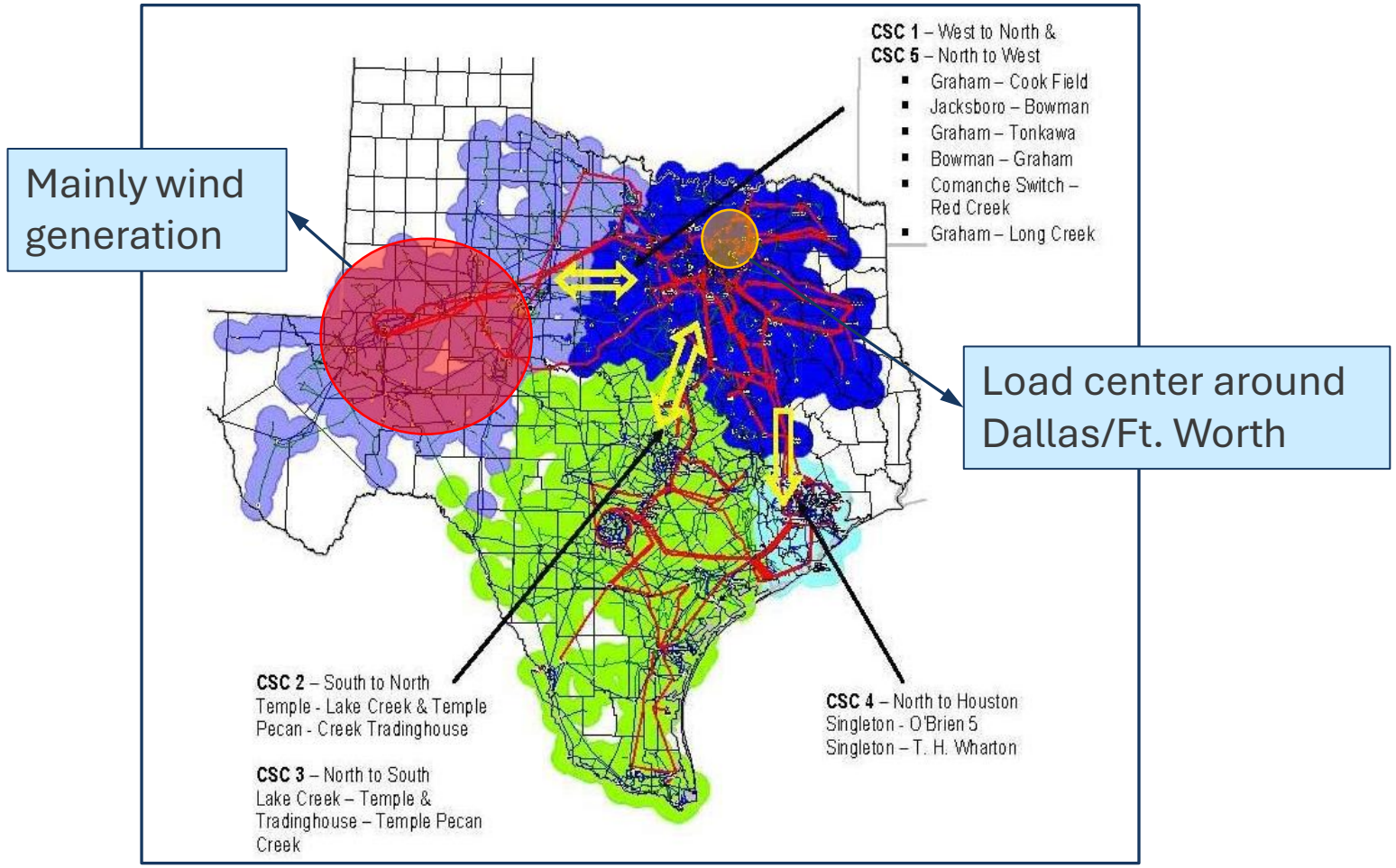
APPLICATION EXAMPLE: ERCOT

- On-line DSA system is based on Powertech's DSATools software
- On-line VSA system
  - Day ahead stability limits posted on Web for use by power market
  - Real-time stability limits computed every hour (8 interfaces) for enforcement in system dispatches
- On-line TSA system
  - Real-time stability limits computed every hour (1 interface) for enforcement in system dispatches
- Study Mode
- Adding more IROL interfaces to On-line DSA



APPLICATION EXAMPLE: ERCOT (CONT'D)

- West-north transmission corridor stability limits



APPLICATION EXAMPLE: ERCOT (CONT'D)

- West-north transmission corridor stability limits
  - With the variability and uncertainty of wind generation, the stability limits calculated using off-line planning models tend to be conservative
  - The limits calculated using on-line TSA could be higher than the planning limits
  - This allows much higher wind generation level in the western region; thus, reducing congestion management cost
  - ERCOT estimated that the use of on-line DSA technology saved **US$27 million** in 2011 on congestion management cost

DSA SYSTEM DEMONSTRATION

IN CASE HELP IS NEEDED…

- **Technical support**
  - For further assistance, contact Powertech at dsainfo@powertechlabs.com

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    Surrey, BC

    Canada, V3W 7R7

- **DSATools™ Web**

  www.dsatools.com